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File Name
03-research-methodology.pdf.pdf

File Size
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1 Research Method and Design

1.1 Introduction

This chapter specifies the research design and methodological framework for the doctoral study Optimizing Software Defect Management, Costs and Operational Efficiency Using AI-Driven Root Cause Analysis and Automated Triage in Micro-service Environments. The study will produce a single primary research artifact — the Heuristic Learning for Integrated Observability Systems framework (HELIOS) — whose construction, ablation, and empirical evaluation will constitute the central contribution of the dissertation. Because the intended output is a purposeful, novel artifact rather than a theoretical generalization about a pre-existing phenomenon, the methodological foundation is Design Science Research (DSR), in the formulation articulated by (Hevner et al., 2004), (Hevner, 2007), and (Peffers et al., 2007). The methodological apparatus comprises three interlocking elements: a pre-registered ablation study of eight confirmatory variants exercised across an AIOpsLab-grounded benchmark corpus with cross-benchmark probes, against eight ablation hypotheses (A-H1 through A-H8) under Holm–Bonferroni family-wise error control with pre-registered rank ordering; a pre-registered baseline-comparison evaluation against five priority-ranked AIOps systems on eight hypotheses (B-H1 through B-H8) corrected as a separate Holm–Bonferroni family; and ten exploratory hypotheses (E-H1 through E-H10) that will generate descriptive evidence informing future confirmatory work without entering the confirmatory hypothesis families.

The chapter is organized as follows. Section 3.2 positions the research philosophically. Section 3.3 specifies the research design, the DSR mapping, the intervention, the unit of analysis, the time horizon, and the construct-to-variable operationalization. Section 3.4 describes the research

setting. A business evaluation layer then maps the technical RCA and triage outcomes onto organizational value after deployment, and specifies the managerial use, the staged adoption path, and the decision rights that go with that deployment. Section 3.5 presents the population and sampling strategy. Section 3.6 specifies HELIOS as the experimental artifact with six named contributions (C1 through C6) decomposed into architectural, algorithmic, and methodological-discipline categories. Section 3.7 describes data-collection procedures, including the exploratory human evaluation that addresses Research Question 4. Section 3.8 specifies the data-analysis plan, and Sections 3.9 through 3.11 discuss validity and reliability, ethical considerations, and methodological limitations.

1.2 Research Philosophy

The philosophical stance of this research is critical realism, tempered by a pragmatist orientation toward artifact evaluation. Critical realism, as developed by Bhaskar (2008/1975) and applied in information-systems research by (Mingers, 2004), holds that reality is stratified into empirical, actual, and real domains, and that generative mechanisms in the real domain produce events in the actual domain, only some of which are observed in the empirical domain. This stance is well suited to the present research domain because microservice failures are produced by real, structural generative mechanisms — fault propagation through typed dependency chains, resource-exhaustion cascades across shared infrastructure, agent reasoning failures in the cognitive substrate — that yield observable symptoms in telemetry streams, yet no single observation constitutes a complete picture of the underlying cause. The epistemic task of a root-cause-analysis (RCA) system is precisely to infer the structural mechanism from the partial telemetry trace it has available; HELIOS will be built on the realist premise that such inference is meaningful because the mechanisms

exist, and on the epistemological premise that such inference is fallible because the mechanisms are only partially observable.

Within this realist frame, the evaluation of HELIOS will adopt a pragmatist stance following the DSR tradition (Hevner et al., 2004; Peffers et al., 2007). The validity of the artifact is not established through correspondence with a pre-existing theory or through the successful prediction of a natural-kind phenomenon; it is established through the artifact's demonstrated utility in solving a specified design problem, measured against pre-registered hypotheses on pre-specified metrics using pre-specified statistical procedures. This pragmatist commitment justifies the emphasis throughout the chapter on falsifiability at the level of each named contribution: every contribution C1 through C6 is ablatable by a feature flag, every ablation corresponds to a hypothesis, and every confirmatory hypothesis is powered on a pre-registered minimum-detectable-effect target.

Two implications follow. The research will not claim that the mechanisms HELIOS infers constitute a complete characterization of microservice faults; it will claim only that HELIOS, under the pre-registered evaluation protocol, will produce inferences that are more accurate, faster, or cheaper than those produced by comparator systems under identical inputs. Nor will the research claim that the architecture of HELIOS is optimal; it will claim only that the architecture is ablatable, that the ablations will be informative about the mechanism of each contribution, and that the pattern of ablation evidence will be consistent with the hypotheses advanced. This modest epistemic posture is appropriate to a doctoral research artifact and protects against the overclaiming that has been a persistent critique in the AIOps sub-field (Pham et al., 2024b; T. Wang & Qi, 2024).

1.3 Research Design

1.3.1 Research Design Classification

The study is classified as Design Science Research with embedded empirical ablation and controlled quantitative comparison. It is not a case study, not an exploratory qualitative inquiry, and not a traditional hypothesis-confirmation experiment on a pre-existing system. It is a construction-and-evaluation study in which the construction of the artifact is itself a research contribution and the evaluation will be structured to falsify, at the level of each architectural mechanism, the claim that the mechanism contributes to the artifact's overall performance. The design integrates three empirical elements: an internal-mechanism ablation study isolating each of the five core contributions of HELIOS (C2–C6); an external comparative evaluation against five priority-ranked AIOps baselines; and descriptive reporting of exploratory variants that cannot, for well-specified reasons, enter the confirmatory hypothesis family under the pre-registration discipline. The methodological consequences of this classification — that the unit of analysis is the paired incident (Section 3.3.4), that the time horizon is cross-sectional with respect to the corpus and longitudinal with respect to ORAR's two-stage reward (Section 3.3.5), and that several mechanisms cannot be confirmatory validated on the benchmark corpus as shipped (Section 3.6.8) — are developed in the corresponding subsections.

1.3.2 Design Science Research (DSR) Framework

The research is organized around the three-cycle DSR framework of (Hevner, 2007). The relevance cycle is instantiated through the dissertation's motivating context: the economic and operational cost of manual triage in cloud-native microservice environments and the emergent class

of agentic-AI failure modes for which no current AIOps framework provides first-class diagnostic support. The rigor cycle is instantiated through the systematic literature review in Chapter Two, through the explicit differentiation of each HELIOS contribution from its most proximate comparator (CHASE for UEG-C, mABC for MAHC, FrugalGPT for ORAR, Istio traffic mirroring for Fidelity-Gated Shadow Validation (FGSV), integrated control-plane self-monitoring patterns for IOL), and through the contribution-back-to-knowledge-base of the runtime-enforced evaluation-rigor discipline (C1) and the five design principles specified in Section 3.6.2. The (Peffers et al., 2007) six-activity DSR process is mapped to the dissertation as follows: problem identification and motivation in Chapters One and Two; objectives in Section 3.8.1 and the pre-registered hypotheses; design and development in the architecture chapter and Section 3.6; demonstration in the pre-registered ablation protocol (eight confirmatory variants × five benchmarks × 40 faults × 10 seeds = 16,000 planned runs) and the 174-fault baseline-comparison protocol; evaluation through the statistical procedures specified in Section 3.8.3; and communication through the dissertation and the Open Science Framework (OSF) pre-registration that will precede and govern data collection.

DSR evaluation rigor will be operationally enforced through architectural commitments that constitute Contribution C1 (Section 3.6.8). First, a Variant Control Layer (VCL) will read a content-hashed variant manifest at process start and emit an immutable feature-flag vector that gates runtime control paths; a flag flip changes which pipelines execute, which consensus rule fires, and which routing arm is selected. Second, a snapshot hash registry will record a content hash of every Unified Evidence Graph with Cognitive Nodes (UEG-C) snapshot and will be queried by downstream pipelines to confirm consumption of the declared snapshot. These two components — specified as Class 1 components of the Evaluation plane (runtime-affecting), as distinct from

Class 2 components (observation-only) — together with the metric integrity gate, the exclusion ledger, the deviation log, and the reconciliation ledger, constitute a research-method-as-system-architecture contribution that converts DSR evaluation rigor from a reporting convention into a runtime invariant of the artifact. Operational enforcement substantially strengthens, but does not by itself dissolve, every threat to paired-observation validity (see Section 3.9.1).

1.3.3 The Intervention Strategy

The intervention is the HELIOS artifact itself, deployed as a treatment in a comparative evaluation design. HELIOS will intervene in the RCA process by replacing existing monolithic or sequential diagnostic pipelines with a peer-pipeline architecture in which deterministic statistical reasoning (D-pipe), graph machine-learning reasoning (G-pipe), and multi-agent large language model reasoning (L-pipe) execute concurrently on the same content-hashed UEG-C snapshot and emit typed verdicts that are fused by the Multi-Agent Hybrid Consensus (MAHC) mechanism. The intervention is not the introduction of any single reasoning modality (each has precedent in the literature) but the structural co-deployment of all three under identical inputs with a cost-aware Operational-Reality-Aware Adaptive Routing (ORAR) mechanism that will select among them based on incident features, per-pipeline historical accuracy priors, and a pre-routing Action-Class Predictor (ACP). What changes under the intervention is the granularity at which RCA reasoning is adjudicated: comparator systems commit to a single architectural reasoning identity, whereas HELIOS shifts the reasoning choice to runtime under cost and latency constraints, with MAHC reconciling verdicts when multiple pipelines execute and the Chain-of-Evidence (CoE) ledger rendering the selection auditable.

The intervention will be executed under six strict control commitments that together constitute the failure-as-covariate discipline and the static-policy commitment:

- **Single-binary variant production**. All eight ablation variants will be produced from a single HELIOS binary (commit SHA fixed at OSF freeze) via feature-flag toggles; no variant is a separate code base.
- **Snapshot identity across variants**. The same UEG-C snapshot, identified by content hash, will be consumed by every active pipeline for the same incident–seed pair.
- **Seed register**. The pre-registered seed register (seeds 10,000,000 through 10,009,999) is binding.
- **Metric integrity gate**. Incident runs with missing measurements will be rejected rather than zero-imputed; excluded runs will be recorded to a cryptographically-signed, append-only exclusion ledger.
- **Price-book freeze**. Per-token and per-compute-second cost coefficients will be frozen at pre-registration; post-freeze price changes will be ignored for confirmatory analysis and noted as an external-validity threat.
- **Static-policy commitment**. The ORAR LinUCB per-arm parameters (A_a, b_a) for a □ {D, G, L, D+G, D+L, G+L, Full} and the Action-Class Predictor weights will be frozen at a pre-warmed state for the entire confirmatory protocol, so that the k-th incident's routing decision is a deterministic function of incident features and frozen parameters and is not conditional on the preceding k−1 incidents' outcomes. This static-policy commitment will preserve the within-pair independence assumption of the paired-comparison design (see Section 3.6.5).

The pre-warmed state will be constructed from a dedicated pre-training corpus that is disjoint from the 174-fault confirmatory corpus and from the 16,000-run ablation corpus. The pre-

training corpus will comprise 1,200 synthetic incidents generated by Chaos Mesh on an isolated three-service synthetic microservice testbed (one stateful database service, one stateless API service, one client-emulator service) that does not appear in any of the five evaluation benchmarks (AIOpsLab, DeathStarBench, Train-Ticket, PetShop, OpsEval) and that is not derived from any of them. The deliberately reduced topology is chosen so that the router will observe fault-class transitions rather than benchmark-specific (fault, service) instances. The synthetic testbed, its fault-injection script, the seed sequence governing the pre-training pass, and the resulting parameter artifacts will be deposited at OSF as a tagged immutable release; their content hashes will be recorded in the same registry that hashes the confirmatory corpus, so that a SHA-256 intersection check at the (fault-type, service-family, injection-parameter) granularity will be executed at OSF freeze. A non-null intersection at freeze would vacate the static-policy commitment, and the affected portion of the confirmatory protocol would be moved to exploratory status before any analysis is conducted. The Variant Control Layer will verify the hash of each loaded parameter artifact against the hash recorded at OSF freeze at every confirmatory run start; hash divergence will block execution.

A second disclosure level addresses the construction protocol itself, distinct from the contents-level disclosure above. The fault-class proportions, severity distribution, and injection-parameter ranges that govern generation of the 1,200-incident pre-training corpus will be pre-registered at OSF prior to corpus generation, and a SHA-256 hash of the construction script — together with the random seed sequence governing fault sampling on the synthetic testbed — will be recorded at freeze and re-verified at any corpus regeneration. This protects against post-hoc selection of construction parameters that would favor any particular routing arm of the LinUCB policy. The chapter acknowledges as a residual construction-validity threat that pre-registration of construction parameters does not, on its own, guarantee that the parameter choices were

neutral with respect to HELIOS's architecture, since the synthetic testbed and the construction script were designed by the same researcher who designed HELIOS; external replication of the pre-warming pass on an independently constructed synthetic testbed is identified in Section 3.11.5 as a priority direction for follow-on research, and is the only intervention that fully neutralizes the construction-validity threat at the process level.

A residual structural-similarity threat is also acknowledged. Even with hash-verified disjointness at the (fault-type, service-family, injection-parameter) granularity, structural similarities between the synthetic pre-training testbed and the AIOpsLab benchmark family — for example, shared HTTP request-response semantics, shared OTEL instrumentation conventions, or shared anomaly-class taxonomy at the conceptual level — could in principle transmit information from the pre-training distribution to the confirmatory distribution through the routing policy, even where no specific fault instance is shared. This residual exposure is bounded by the small dimensionality of the LinUCB feature space (approximately 12 to 18 features comprising incident features and the ACP stateful-action feature block) relative to the diversity of the confirmatory corpus, and by the deliberate construction of the synthetic testbed to expose the router to fault-class transitions rather than to specific instances; neither bound is a guarantee. The threat is reported in Section 3.9.1 (Threat 8) and Section 3.11.5; the chapter identifies external replication on a structurally distinct microservice paradigm — for example, an event-driven serverless mesh in place of a request-response Kubernetes deployment — as the only intervention that fully neutralizes the threat at the process level, and does not claim to dissolve it through the present disclosure regime alone.

1.3.4 Unit of Analysis

The primary unit of analysis is the paired incident instance: a single fault injection, executed under a specified seed, producing paired observations on HELIOS-Full and on each comparator variant or baseline system under identical telemetry access. Pairing at the incident level is both statistically and philosophically necessary. Statistically, paired observations reduce variance attributable to fault heterogeneity and admit the non-parametric Wilcoxon signed-rank test as the primary inferential instrument; the power analysis grounded in CHASE's reported HR@3 performance (Cohen's h = 0.276; derivation in Section 3.8.3) yields a minimum required n of 73 paired faults at Holm–Bonferroni $\alpha = 0.00625$ and 80 percent power, inflated to n = 84 under a 15 percent attrition allowance, which justifies the planned 174-fault delivered corpus at 2.1× the attrition-inflated minimum (see Section 3.5 and Section 3.8.3). The validity of the paired design relies on within-pair independence of treatment effects across incidents, which is operationally protected by the static-policy commitment in Section 3.3.3. Philosophically, the paired-incident unit respects the critical-realist premise that each incident is a distinct generative event, and the pragmatist premise that each incident is a distinct problem instance on which the artifact's utility will be measured.

Two secondary units supplement the primary unit. The first is the variant–benchmark–metric cell (the atomic unit of confirmatory inference), with eight confirmatory variants × five benchmarks × five confirmatory metrics yielding 200 cells in the pre-registered results matrix. A cell will be eligible for confirmatory inference only if at least 320 of 400 planned paired observations (80 percent) are included under the inclusion criteria specified in the pre-registration; cells below this threshold will be reported as "insufficient data" with the completed count and the reason,

and they will not enter the hypothesis tests. The second supplementary unit is the contribution–ablation pair: the binding of each named core contribution (C2–C6) to its designated primary ablation variant under the discipline specified in Section 3.6.7 and Section 3.6.8.

1.3.5 Time Horizon

The primary time horizon of the empirical evaluation will be cross-sectional. The benchmark corpus will be fixed at pre-registration freeze (with SHA-256 hashes published alongside the reserve-pool hashes for replacement instances), and all variants will be evaluated against the same corpus under seed-controlled replication. Each of the five benchmarks will contribute a pre-specified set of fault instances drawn from the fault categories documented in the pre-registration appendix; no benchmark result will be aggregated across benchmarks without explicit per-benchmark reporting alongside.

A secondary longitudinal element exists within the ORAR routing mechanism. The router will emit a provisional reward r□¹□ at approximately one to five minutes after actuation and a terminal reward r□²□ when the post-mortem label arrives (hours to days later). The Reward Reconciliation Error (RRE) metric is defined as the discrepancy between the sign of the provisional reward and the sign of the reconciled terminal reward. Because the benchmark corpus does not produce naturally delayed labels (ground-truth root-cause is available at the chaos-mesh injection time, so provisional and terminal rewards collapse to the same instant), and because the LinUCB parameters will be frozen for the confirmatory protocol under the static-policy commitment (Section 3.3.3), neither the confirmatory RRE analysis nor the on-line ORAR learning dynamics will be evaluated on the confirmatory corpus. The RRE measurement gap will be addressed through synthetic delayed-label injection on a pre-registered 25 percent subset of AIOpsLab and Train-Ticket

incidents, parameterized as a log-normal distribution with $\mu_{log} = 8.7$ (in seconds) and $\sigma_{log} = 1.4$, yielding a median delay of approximately 1.7 hours, a 25th-percentile delay of 38 minutes, a 75th-percentile delay of 4.6 hours, and a 95th-percentile delay of approximately 19 hours. This parameterization reflects the 'hours to days' production-incident reality that the RRE metric is designed to characterize and is grounded in the post-mortem arrival distributions reported in (Pham et al., 2024a) and in the cross-platform retrospective summarized in (T. Wang & Qi, 2024). ORAR on this 25 percent subset will operate with thawed LinUCB updates and will be reported as a disjoint exploratory analysis; the RRE metric and the on-line learning behavior will be reported only on these exploratory subsets and will not constitute confirmatory tests.

1.3.6 Variables and Constructs

The research operationalizes six primary constructs corresponding to Contributions C1 through C6, each mapped to an architectural mechanism of HELIOS and each measured through pre-specified metrics. The constructs, their architectural source, their independent-variable and dependent-variable operationalization, and their measurement instrumentation are summarized in following Table.

The construct-to-component binding is itself an instance of Contribution C1: each independent variable corresponds to a feature flag (or, for C1, to an architectural commitment) and each dependent variable is recorded at a specific architectural location, so the operational definition of every construct is fixed at runtime by the architecture rather than at analysis time by the researcher. This architectural fixing is a reliability commitment — it eliminates the researcher-degrees-of-freedom risk that the same nominal construct could be measured at two different architectural locations producing two different operational definitions — and is in this respect a positive

contribution to the chapter's methodological discipline. The chapter does not, however, claim that architectural fixing enforces construct validity in the stronger sense of guaranteeing alignment between the operational measurement and the theoretical construct it is intended to reflect. The architectural binding strengthens the reliability of construct measurement (every measurement of construct X will reflect the same operationalization across runs and variants) without claiming to strengthen the validity of that operationalization as a reflection of the underlying theoretical construct.

Table 1
Constructs, Variables, and Operationalization in HELIOS Evaluation

Construct	Architectural Source	Independent Variable	Dependent Variable	Measurement Location
Runtime-enforced evaluation rigor	C1 — Variant Control Layer + snapshot hash registry + ledger apparatus	Architectural commitment; not feature-flag-toggled	Variant-identity verification rate; snapshot-consumption verification rate; metric-integrity-gate completion rate; deviation-log entry count	Evaluation plane; runtime hash-verification logs

Construct	Architectural Source	Independent Variable	Dependent Variable	Measurement Location
Cross-domain evidence integration	C2 — UEG-C with cognitive nodes	`p4_cognitive` flag (L-pipe scope)	HR@3 on agentic-AI fault classes	Plane 4 MAHC ranked output
Shadow validation fidelity	C3′ — FGSV with KS gate	`fgsv_ks_gate` flag	False-accept rate of invalid actions	L4 FGSV verdict record
Uncertainty-aware consensus	C4 — MAHC with hierarchical priors	`mahc_hierarchical` flag	HR@3; consensus-confusion rate	Plane 4 MAHC ledger
Meta-observability	C5 — IOL self-telemetry	`iol_self_telemetry` flag	Degradation detection latency	L1 `tenant=self` partition
Cost-aware adaptive routing	C6 — ORAR (Contextual Bandit Router (CBR) + ACP + reconciliation)	`cbr_routing`, `acp_stateful`, `reconcile_terminal`, `his_tier` flags	CpR at matched HR@3; RRE; A1 latency per HIS tier	ORAR arm-selection log; reconciliation ledger

Note. HR@3 = hit rate at 3; CpR = cost per resolution; RRE = Reward Reconciliation Error; HIS = hardware-isolation tier; ACP = Action-Class Predictor. With the exception of C1 (which is an architectural commitment to evaluation-rigor infrastructure rather than a feature-flag-toggled mechanism), each independent variable is a feature flag toggled at the Variant Control Layer. Each

dependent variable is recorded at a specific architectural location and cannot be measured elsewhere without violating the integrity of the paired observation.

A specific instance of the architectural-fixing distinction warrants explicit treatment. HR@3 will be measured at the MAHC ranked-output module in HELIOS-Full. In HELIOS-noConsensus, MAHC will be ablated and replaced by a vote-only aggregator that emits a structurally identical ranked-output tuple at the same Plane 4 architectural location under the same content-addressable schema. The measurement location is therefore architecturally identical across HELIOS-Full and HELIOS-noConsensus, but the mechanism producing the measurement differs (uncertainty-aware hierarchical consensus in Full; scalar-vote aggregation in noConsensus). The chapter accepts that this is a single-construct operationalization rather than two distinct constructs, and that A-H4 (no-consensus degrades HR@3 on ambiguous incidents) is a comparison of the same construct under two consensus mechanisms. The construct-validity asymmetry that this resolution does not dissolve — that the noConsensus measurement reflects a different aggregation mechanism even though it occurs at the same architectural location — is acknowledged in Section 3.9.2 and is treated there as a residual threat that the architectural-fixing argument bounds but does not eliminate.

Beyond the five core constructs, three secondary constructs will be measured descriptively but will not enter the confirmatory hypothesis set: hallucination rate, decomposed into factual and structural components (H_fact and H_struct) with independent-rater labeling against pre-specified rubrics; time-to-triage (TTT), measured as the interval between the anomaly sentinel event timestamp and the publication of the MAHC provisional verdict; and SRE interpretation quality, to be measured in the exploratory HELIOS-noExpl variant for the user study accompanying Research Question 4.

1.4 Research Setting and Context

1.4.1 The Architectural Setting

The research is situated in cloud-native microservice environments orchestrated on Kubernetes and instrumented with OpenTelemetry (OTEL)-compliant observability tooling. The HELIOS artifact will enforce a strict physical separation between the Data Plane, which hosts production microservices, serverless functions, and agentic AI applications, and the Control Plane, which hosts HELIOS itself. The Control Plane will execute on a dedicated Kubernetes node pool tainted against Data Plane workloads, with ResourceQuota caps of eight CPU cores and 32 GiB of memory per incident, and OTEL collectors will be deployed as DaemonSets on Data Plane nodes with CPU limits of 100 millicores per node, ensuring that observability instrumentation cannot directly exacerbate resource-exhaustion incidents. HELIOS will be organized as a five-layer execution hierarchy (L0 telemetry ingress; L1 UEG-C construction; L2 peer reasoning pipelines; L3 coordination and consensus; L4 action and learning) orthogonal to five governance planes (Data, Reasoning, Coordination, Learning, Evaluation), with the Evaluation plane spanning all layers; the per-component specification is given in Section 3.6.3. The architectural setting is a methodological precondition for causal identifiability of each architectural contribution under the bounded scope conditions of the study: peer-pipeline co-deployment on identical inputs is the minimum structural commitment that turns the question “what does LLM reasoning contribute to RCA?” into a tractable question, because HELIOS-noLLM will leave the D-pipe and G-pipe as a functioning residual system against which HELIOS-Full can be compared on identical inputs.

1.4.2 The Business and Operational Context: High-Stakes Reliability

The operational context is characterized by the economic and reputational consequences of service degradation at cloud scale, with industrial incident reports documenting outage costs reaching tens of millions of dollars per hour in high-traffic financial-sector and e-commerce deployments (Pham et al., 2024a). The research addresses environments in which service-level objective (SLO) breach penalties are contractually specified, multiple tenants with heterogeneous service-class requirements share cluster infrastructure, and the cost of large language model inference for AI-driven diagnosis has itself become a material operational expense that must be architecturally managed rather than treated as a parameter to be minimized post hoc. The context shapes the research design in three ways: economic proportionality is operationalized through the Cost per Resolution (CpR) metric (Hypothesis A-H5 of the ablation family and Hypothesis B-H7 of the baseline-comparison family) as a technical proxy for organizational outcomes (on-call burden, customer-facing recovery, cloud spend) grounded in industrial retrospectives; safety is operationalized through the FGSV shadow validation mechanism and the Human-in-the-Loop (HITL) gate stratified by dynamic autonomic threshold α; and adversarial robustness is operationalized through the Adversarial Robustness Monitor that inspects incoming telemetry for prompt-injection payloads and verifies agent integrity via periodic attestation challenges. These three operational emphases are architectural commitments required by the high-stakes context and will be ablatable through the FGSV, HITL, and Adversarial Robustness Monitor (ARM) feature flags respectively (Section 3.6.7). The proxy nature of the technical metrics is acknowledged as an external-validity boundary in Section 3.11.2.

1.4.3 The Data Context: The Three Pillars of Observability

The research is grounded in the three canonical observability pillars established in cloud-native practice: metrics (P1, collected via Prometheus exporters and OTEL counters), distributed traces (P2, via Jaeger and OTEL with standard semantic conventions), and logs (P3, parsed via Drain3 template extraction (Duan et al., 2024) with LogEDL evidential deep learning for vacuity estimation). HELIOS extends this canonical framework with two additional pillars specified as first-class signals rather than optional supplementary data: cognitive telemetry (P4 — agent reasoning-step spans under the Model Context Protocol convention, tool-call schema-validation records, goal-drift embedding cosine-distance measurements, and loop-depth counters) and business telemetry (P5 — transaction-level revenue signals, SLO breach penalty coefficients, and service-class tier assignments).

HELIOS will consume all five pillars through a single UEG-C representation with four typed edge classes (structural, behavioral, causal, economic) and formal composition rules, such that cross-domain path queries — causal composed with economic, behavioral composed with structural — will be first-class operations on the graph. This integration distinguishes HELIOS from prior single-pillar systems (Liu, 2021; Pham et al., 2024b) and from prior multi-modal systems that maintain pillar-specific pipelines emitting separate verdicts (Yu et al., 2023; Zhao et al., 2025; Zheng et al., 2024). The five-pillar extension is presented as a conceptual-architectural contribution rather than a strict ontological claim that P4 and P5 must be separate pillars; the substantive content is the integrated UEG-C treatment with formal composition rules across the five signal classes. Whether the integration is the source of HELIOS's empirical performance advantage, or whether the

advantage is carried by the pipeline architecture alone, is the question that the HELIOS-noStructural ablation is designed to probe under hypothesis A-H8 (Section 3.6.7).

1.4.4 The Technological Setting: The Emergence of AIOps

The research is situated at the intersection of three technological trajectories: the maturation of Kubernetes as the de facto microservice orchestration standard, the proliferation of agentic AI systems built on large language models, and the emergence of AIOps as a research and industrial discipline. The AIOps landscape surveyed in Chapter Two encompasses a growing body of academic systems for microservice RCA and a complementary body of commercial platforms providing anomaly detection enriched with varying degrees of LLM-assisted explanation. A systematic gap persists across the surveyed landscape: no existing system simultaneously satisfies five methodological desiderata necessary for causally identifiable evaluation of AI-driven RCA:

- **D1:** Peer-pipeline co-deployment of statistical, graph-ML, and LLM reasoning on identical inputs.
- **D2:** First-class observability for LLM agent reasoning failures alongside infrastructure faults.
- **D3:** Economic arbitration of diagnostic effort through cost-aware routing grounded in observed accuracy priors.
- **D4:** Deferred two-stage reward reconciliation that accounts for the asynchronous arrival of post-mortem labels.
- **D5:** Ablation-first architectural discipline that produces variant equivalence at the binary level rather than through retrained model weights.

Table 2
Comparative Analysis

System	D1 Peer pipelines on one input	D2 Own reasoning observed	D3 Cost routed by accuracy	D4 Two-stage reward	D5 Same-build ablation
CHASE	No	No	No	No	No
mABC	Partial	No	No	No	No
TAMO	No	No	No	No	No
Flow-of-Action	No	No	No	No	No
FrugalGPT	No	No	No	No	No
HELIOS	Proposed	Proposed	Proposed	Proposed	Proposed

The desiderata are derived from examination of methodological gaps in the comparator corpus rather than from the HELIOS specification itself; HELIOS is one possible artifact that will satisfy all five, with D1 corresponding to Contribution C1 (peer-pipeline integration enforced by C1's runtime apparatus), D2 to C2, D3 and D4 to C6, and D5 to C1's ablation-first architectural discipline.

The C5 (meta-observability) contribution further differentiates HELIOS from concrete control-plane self-monitoring patterns — Prometheus self-scrape, OpenTelemetry self-instrumentation, and the meta-monitoring features of commercial observability platforms such as Datadog and New Relic — through the integrated-graph treatment of self-telemetry as a tenant=self partition in the same UEG-C that the production-diagnostic pipeline consumes

(Section 3.6.8). Because each comparator baseline satisfies some but not all of the five desiderata, cross-comparator generalizations will be interpretive rather than strictly inferential; the analysis plan in Section 3.8 therefore treats the baseline-comparison family as a separate hypothesis family from the ablation family, corrected separately under Holm–Bonferroni to avoid combining incommensurable inference targets into a single family-wise error budget.

1.5 Business Evaluation Layer

A DBA dissertation that stops at hit-rate and latency has not finished the methods argument. HELIOS is meant to sit in a production estate after the evaluation, and the organization that would pay for it does not buy HR@3. It buys shorter incidents, cheaper resolution, fewer SLO credits written, and diagnoses that people will actually act on. This section is the methods account of that translation. It is not a new confirmatory family. The laboratory still measures the technical variables in Section 3.3.6 and Section 3.8.4. What follows is how those variables are read as business outcomes, what the architecture already records for that reading, and what remains a post-deployment claim rather than a result of the 16,000-run protocol.

The mapping is one-way and lossy, which is worth saying before the table. A drop in log-MTTR on a chaos-injected fault is not the same thing as recovered revenue on a shopping day. CpR is assembled from token spend, infrastructure, storage, human minutes, and opportunity cost at the Critic Agent; it is a cost object, not a P&L line. The chapter already treats these metrics as proxies (Section 3.4.2). The business layer does not pretend otherwise. It specifies which proxy stands for which organizational effect, so that Chapter Four can report the technical tests without leaving the business reading implicit, and so that a later industrial deployment has a documented path from those tests to value.

Table 3
Technical outcomes to business outcomes after deployment

What HELIOS measures	Hypothesis / instrument	Business reading after deployment	Who sees it
HR@3, Avg@5	A-H1–A-H4, A-H6, A-H8; B-H1–B-H5, B-H8	Fewer wrong first diagnoses; less time chasing the wrong service; fewer repeat incidents from a missed root cause	SRE lead; incident commander
log-MTTR	B-H6	Shorter customer-visible impact window; less SLO burn; lower contractual service-credit exposure	Service owner; finance
CpR / log-Cost at matched accuracy	A-H5; B-H7	Lower cost per closed incident, including LLM inference, so diagnosis does not become a second opex problem	Platform owner; FinOps

What HELIOS measures	Hypothesis / instrument	Business reading after deployment	Who sees it
Time-to-triage (TTT)	E-H2 (exploratory)	Faster hand-off from alert to a named cause; less on-call queueing	On-call rotation
Correct interpretation, trust, accept / escalate / reject	E-H7 user study	Decision effectiveness: whether the output is usable enough that a person will act, escalate, or stop a bad remediation	SRE; service owner; academic reviewer of the design
Hallucination rates (H_fact, H_struct)	A-H7	Residual risk of a fluent but wrong narrative driving a bad change	Risk / change-advisory
FGSV false-accept; HITL on high-risk actions	E-H4; Monitor Agent	Blast-radius control; the organization is not buying unsupervised writes to production	Engineering leadership

Four of those rows are the ones DDR is asking for, in ordinary language.

Operational efficiency is MTTR and TTT. If HELIOS-Full is faster to a usable diagnosis than the best baseline (B-H6) and faster to a provisional verdict than uniform routing (E-H2), the

operational story after deployment is fewer engineer-hours per incident and a shorter wait before someone can start a fix. Published deployments in this literature have claimed diagnosis moving from hours to minutes (Chakraborty et al., 2024; Guntupalli, 2025). Those figures are not HELIOS results. They are the class of benefit the MTTR and TTT instruments are built to speak to, once the system is on a live estate.

Incident-resolution cost is CpR. A-H5 asks whether turning ORAR off makes that cost worse at the same accuracy; B-H7 asks whether HELIOS-Full is cheaper than the best baseline. After deployment the same object is what FinOps would recognize: tokens, cluster time, storage, and the human minutes still spent on the ticket. The design commitment in Chapter Two was that generative RCA can be accurate and still too expensive to run on every alert. ORAR exists because of that. The business layer therefore treats a CpR reduction at matched HR@3 as the organizational value of C6, not as a side-effect.

Service availability and impact are not measured as uptime percentages in the confirmatory corpus, because the corpus is injected faults with a known start, not a year of production traffic. The available proxy is the duration of the diagnostic and recovery path (log-MTTR) together with HR@3 (whether the recovery path is aimed at the right component). P5 business telemetry — revenue signals, SLO penalty coefficients, service-class tiers — and the economic edge type in UEG-C are how HELIOS would consume those signals if they exist on a live estate. In the laboratory they are specified so that a later deployment can attach a dollar figure to an SLO burn without changing the diagnostic core. Recovered-revenue numbers will not be reported from chaos-mesh runs.

Decision effectiveness is the RQ4 study and the HITL log. Accuracy on a ranked list is necessary and not sufficient. If the Chain-of-Evidence form is no easier to read than the raw list, the organization has bought a faster wrong conversation. E-H7 is exploratory and n = 24 is mixed

(practitioners, academics, business participants) for that reason: after deployment the output is not only read by the on-call engineer. A service owner who has to approve a rollback, or a reviewer who has to believe the write-up, is part of the value path. The study does not estimate adoption in the field. It asks whether the narrative is usable enough to change interpretation, trust, and the accept / escalate / reject choice.

1.5.1 Managerial application

The table above names who reads each number. The rest of this subsection says what they are deciding. The incident process keeps its ordinary shape: detect, diagnose, decide, remediate, learn. The architecture in Section 3.6.3 is that sequence drawn as boxes.

Detection is L0. Intake takes the telemetry the estate already emits — metrics, traces, and logs — and puts two further signals on the same incident: what the firm's own agents were doing, and the business weight of the service (revenue, SLO penalty, service tier). Which services are in that intake is settled by the platform owner and the service owners. HELIOS does not draw the estate boundary for them.

Diagnosis holds the two decisions that change the cost of the incident. The ranked list is which service to open first. A wrong first guess is paid for by the SRE lead and the incident commander, which is why they sit on the HR@3 and Avg@5 row. The router is the other decision: whether this alert earns a language-model call, or whether the cheaper pipelines are enough. The platform owner and FinOps already know that object as cost per resolution. A-H5 and B-H7 are the tests. Until they are run, the managerial claim is only that the cost is written down in parts those roles recognize: tokens, cluster time, storage, human minutes, and opportunity cost.

The decide step is the evidence trail out of L3. A ranked list that nobody will sign does not shorten the incident. It shortens the argument and then leaves the argument where it was. E-H7 asks whether a person can read the trail and accept it, escalate it, or reject it. After deployment that person is the on-call engineer, and also the service owner who has to allow a rollback, and the reviewer who has to stand behind the write-up. They are already on the decision-effectiveness row.

Remediation is L4, and it is fenced. The shadow test runs before a production write. A database change, a network-policy change, or a rollback waits for a named person. Engineering leadership's stake in that row is the blast radius: the firm is paying for a diagnosis it can refuse. Learning is the outcome record on the same layer. In the confirmatory protocol that record does not move the router's parameters. The update path is there for the exploratory delayed-label subset and for a later deployment. Chapter 4 will not report it as an effect of the 16,000 runs.

1.5.2 Organizational adoption

Whether a firm adopts HELIOS is not a result this study will estimate. Chapter 1 already lists culture and management among the limits on that outcome. H3b is smaller than a deployment claim. If trust in the output rises, stated willingness to accept the diagnosis inside the user study may rise with it. That is a question about the exercise.

What can be specified now is the order in which the controls would be opened. (Rogers, 2003) is the reference for the plain conditions under which an operations team tries a tool: the gain has to be visible, the tool has to fit what the team already runs, and the team has to be able to try it without putting a production write at risk. The three stages below are those conditions, assembled from controls the architecture already has.

Shadow is the trial. HELIOS writes the diagnosis and the cost per resolution, and stops. The human gate stays shut. Nothing is handed to production. A platform team can put the ranked cause, the time, and the cost next to what the on-call engineer actually did.

Recommendation is the mode the study can speak about. Every action waits for a person, and the evidence trail is what they read. E-H7 is this mode. Twenty-four participants — practitioners, academics, and business readers — can show whether the narrative changes interpretation, trust, and the accept / escalate / reject choice. They cannot show that an organization will keep the tool. Assumption A3 already says the same thing in operational terms: findings should not be carried to a firm that has not had a 30-day period of human calibration. That month is this stage. Assumption A5 sits beside it. With a thin incident history, SOP matching stays weak for about 90 days, and early recommendations should be read with that in view.

Automatic action is the firm's option, and it is optional. The firm sets the risk threshold α. Actions under the line may run without a person in the loop. A rollback, a data change, or a network-policy change still waits, and the shadow copy still has to agree with live telemetry before a write. HELIOS does not arrive with α filled in.

The proxies remain proxies. (Kohli & Grover, 2008; Melville et al., 2004) treat an IT measure as a stand-in for an organizational effect, with the effect itself still unobserved on the firm's books. A lower log-MTTR on a chaos-injected fault is still not recovered revenue. Putting a rollout order around the artifact does not change that.

1.5.3 Governance

The five planes in Section 3.6 — Data, Reasoning, Coordination, Learning, and Evaluation — place components. They do not say who is allowed to let an action through. That second question belongs to the firm, and the third stage above is empty until it is answered.

Table 4
Decision rights for a HELIOS deployment

Rule	Who decides	What the artefact already records
Which alerts are in scope	Platform owner, with the service owners	Sampling tier and incident signature
When the language model may be called	FinOps with the SRE lead, through the cost target	Cost per resolution, and the router's choice
Which actions may run unattended	The firm, by setting α	Action class and the threshold
Who approves a high-risk change	A named person, for a rollback, a data change, or a network-policy change	Human gate and the Chain-of-Evidence
When a fix may touch production	The change owner, after the shadow copy agrees with live telemetry	Shadow-validation verdict
What personal or tenant data may reach the model	The data owner	Privacy gateway and tenant separation

Rule	Who decides	What the artefact already records
How a fluent but wrong diagnosis is caught	Risk reviewer	Factual and structural hallucination checks, and the ethics entry on the evidence ledger
Who reviews overrides and spend	SRE lead, platform owner, and risk, on the firm's existing incident cadence	Override, exclusion, and cost records

The person or function that sets α owns the decision that some actions may run unattended. The Chain-of-Evidence is how that decision is reconstructed: which threshold was in force, who was asked, what the shadow test returned, and what was written. An audit trail supports reconstruction. It does not settle legal liability. Who answers for a bad automatic change is outside this study.

Privacy, consent, and the requirement that a high-risk action wait for a person remain in the ethics section. That section is about the research. The table here is about the firm that would run the artefact afterwards. Overrides, exclusions, and cost per resolution are already on the ledger for the SRE lead, the platform owner, and risk to read. The study does not convene that review.

What this layer will and will not claim in Chapter Four follows from the above. Confirmatory tests may support (or fail to support) lower log-MTTR, lower log-Cost, and higher HR@3. Those results, if obtained, license a bounded business reading: under the corpus and the frozen protocol, the artifact is cheaper and/or faster and/or more accurate than the named comparators, which

is the usual mechanism by which RCA tooling earns its keep. They do not license a claim that a named firm will save a named sum, that availability will rise by x nines, or that SRE toil will fall by the 41 percent reported in other settings (Hwang et al., 2021). Those claims need production traffic, a price book for SLO credits, and a time-on-ticket measure that the present instruments only approximate. The adoption stages and the decision-rights table specify how a firm would open the controls and who would set the thresholds. They are not evidence that any firm will. An industrial replication, if one is engaged, is where that last step belongs (Section 3.11.5). Until then the business layer is a documented translation, not a second results chapter.

1.6 Population and Sampling

The target population for the quantitative evaluation is the set of fault scenarios representable in cloud-native microservice environments as exercised through chaos-engineering fault injection and workload-driven degradation. The confirmatory population is operationalized through AIOpsLab (Y. Chen et al., 2025) as the primary benchmark, with cross-benchmark probes drawn from DeathStarBench (Gan et al., 2021), Train-Ticket, PetShop (Hardt et al., 2024), and OpsEval (Liu et al., 2025) supporting external-validity assessment but not confirmatory cross-benchmark inference (see Section 3.9.3). The five benchmarks are selected because they provide publicly available, independently replicable fault dataset with documented annotation protocols (a precondition for reproducibility under the DSR framework), they span heterogeneous microservice architectures (hotel-reservation, social-network, train-ticket, e-commerce, and IT-operations task suites) supporting external validity across application domains, and they exhibit different fault-injection regimes (Chaos Mesh, ChaosBlade, functional fault injection) that prevent the evaluation from being over-fit to the failure mode of a single injection tool.

The sampling strategy combines stratified and purposive elements under the binding constraint of the a priori power analysis. Each benchmark will contribute 40 pre-specified fault instances drawn from the fault categories documented in the pre-registration appendix (resource exhaustion, network partition, service degradation, dependency failure, database contention, and benchmark-specific fault classes). With ten random seeds per fault instance, each variant–benchmark cell will produce 400 paired observations, for a full ablation protocol of eight variants × five benchmarks × 40 faults × 10 seeds = 16,000 incident runs. The 16,000-run ablation protocol will be paired with a 174-fault delivered corpus for the baseline comparison, sized to 2.1× the 84-fault statistical minimum determined by power analysis (Section 3.8.3).

The 174-fault corpus will be constructed across four augmentation tiers to reconcile the conflicting demands of statistical power, external validity, and cognitive-fault coverage. Tier 1 will comprise 96 paired faults drawn from two AIOpsLab replicas under distinct random seeds, integrated in analysis through a generalized linear mixed-effects model (GLMM) with ‘problem_id‘ as a random intercept to prevent pseudoreplication inflation of degrees of freedom. Tier 2 will comprise 36 paired faults from a Chaos-Mesh parametric extension of AIOpsLab (9 fault types × 3 severity levels × 2 concurrency levels in a balanced factorial). Tier 3 will comprise 30 paired faults from Train-Ticket and DeathStarBench under ChaosBlade injection, contributing cross-benchmark external validity. Tier 4 will comprise 12 paired cognitive-agentic synthetic faults constructed through NIKA-style structured Model Context Protocol instrumentation (Z. Wang et al., 2025), comprising six hallucination-class faults (invalid tool schema, non-existent API invocation, argument fabrication, RAG-contamination, self-contradiction, context-leakage) and six goal-drift-class faults (task-objective decay, looping, premature termination, delegation misrouting, persona violation, ethics-bypass attempt). Tier 4 faults will be layered onto a subset of AIOpsLab incidents

through cognitive-agentic instrumentation rather than constructed on an independent synthetic substrate; the cognitive-node telemetry is injected into the AIOpsLab incident harness, and the resulting paired observations therefore inherit AIOpsLab's microservice topology, fault-injection regime, and labeling discipline.

The 174-fault corpus exhibits a substantial concentration in the AIOpsLab benchmark family: Tier 1 and Tier 2 together comprise 132 of 174 instances, or approximately 76 percent of the corpus. The methodological consequence is that confirmatory inferences will be bounded to the class of AIOpsLab and its parametric extension; cross-benchmark results in Tiers 3 and 4 will be reported as external-validity probes that suggest, but do not confirm, that the observed effects extend to environments outside the AIOpsLab specification (see Section 3.9.3). The rank-based mixed-effects treatment of problem_id as a random intercept will control for pseudoreplication within AIOpsLab but does not, on its own, repair the bound on the population to which findings can be generalized.

Tier 4 is not optional within the corpus — the C2 cognitive-agentic claim requires agentic-AI fault instances that no baseline system can diagnose — but two methodological consequences are explicitly accepted: the n = 12 sample size is below the 73-fault statistical minimum, and Tier 4 instances are constructed by the researcher specifically to exercise cognitive-node reasoning. These two consequences together motivate the reclassification of the C2 cognitive-agentic evaluation as exploratory empirical validation under E-H1 rather than confirmatory inference (Section 3.6.8 and Section 3.8.2). Tier 4 evidence supports an existence-proof claim — that HELIOS can diagnose cognitive-agentic faults that comparator baselines cannot — but does not support a population-level efficacy claim about cognitive-agentic fault diagnosis in general; an externally constructed

cognitive-agentic fault corpus on a non-AIOpsLab substrate is identified in Section 3.11.5 as the priority direction for upgrading E-H1 to confirmatory status.

Four statistical-validity safeguards will protect inferential validity under augmentation: ‘benchmark_source‘ will be included as a fixed effect in the GLMM to prevent between-source heterogeneity from inflating Type I error; Tier 1 replicates will share ‘problem_id‘ as a random intercept to avoid pseudoreplication; the GLMM specification, corpus partition, and Holm–Bonferroni hypothesis ordering will be locked at pre-registration freeze and deposited at OSF; and results will be reported at ρ □ {0.4, 0.5, 0.6} to demonstrate robustness to the paired-correlation assumption embedded in the sample-size calculation.

The 174-fault baseline-comparison corpus is a subset of the 200-fault ablation corpus drawn from the AIOpsLab and Train-Ticket benchmarks (132 from Tier 1+2 AIOpsLab, 30 from Tier 3 Train-Ticket and DeathStarBench, and 12 from Tier 4 cognitive-agentic on AIOpsLab); the remaining 26 ablation faults are PetShop and OpsEval instances used in the ablation protocol but not in baseline comparison because the baseline systems do not all support those benchmark substrates. The corpus relationship is documented in the OSF-deposited fault-instance manifest with full SHA-256 hashes for each instance’s membership in each protocol.

The pseudoreplication-control specification deserves explicit binding under the hypothesis schema. To preserve coherence with the non-parametric Wilcoxon primary tests (Section 3.8.3), pseudoreplication on Tier 1 (the 96 paired faults from two AIOpsLab replicas under distinct seeds) will be controlled via a rank-based mixed-effects analysis: the paired outcome differences will be rank-transformed within each (variant, benchmark, metric) cell prior to fitting a linear mixed model with problem_id as a random intercept and benchmark_source as a fixed effect. Variance components will be estimated by restricted maximum likelihood (REML); degrees-of-freedom ad-

justment will be Kenward–Roger; the residual covariance for the seed-paired observations within each problem_id will be unstructured. As a concurrent sensitivity analysis, the non-parametric repeated-measures procedure of (Noguchi et al., 2012) the nparLD package in R) will be applied to the same Tier 1 subset and reported alongside. Neither analysis is the binding confirmatory test for any hypothesis; the binding tests are specified in Section 3.8.3 (Wilcoxon signed-rank for the ablation family; McNemar's exact for B-H1 through B-H5; Wilcoxon signed-rank for B-H6 through B-H8 with paired t on log-transformed values reported as sensitivity analysis). The pseudoreplication-control analyses verify that the Wilcoxon inference does not shift materially when the within-Tier-1 dependence structure is modeled explicitly; if the rank-based mixed model and the Wilcoxon results disagree on direction or significance for any A-Hx, both will be reported and the disagreement treated as a sensitivity-analysis flag rather than as confirmatory inference. The pseudoreplication-control specification will be locked at OSF freeze and any departure recorded in the deviation log.

The sampling is not intended to support population-level generalization to every microservice environment in industry. It is intended to support the narrower and more defensible claim that, within the class of AIOpsLab-representative microservice environments and across the four corpus tiers, HELIOS's architectural contributions will produce the hypothesized effects on the pre-specified metrics at the pre-specified effect sizes, and that cross-benchmark observations in Tiers 3 and 4 will be consistent with — but will not confirm — generalization beyond AIOpsLab and its parametric extension.

1.7 The AI Artifact Development (The Intervention)

1.7.1 Research Artifact Definition

The research artifact is HELIOS, the Heuristic Learning for Integrated Observability Systems framework, specified as a multi-agent, multi-modal, ablation-first RCA system for cloud-native microservice environments that co-exist with agentic AI applications. HELIOS's definitional scope is bounded by six explicit commitments: it will operate on OTEL-instrumented microservice estates of at most approximately 100 services within the benchmark regime; it will assume an enterprise-scale control-plane compute budget of 5 to 10 percent of cluster capacity; it will require seeded accuracy-calibration data from pre-production fault-injection campaigns; it will assume LLM-access tier 2 or higher (logprob-API access) for full P4 cognitive telemetry; it will assume the existence of historical incident records for the Knowledge Cortex warm tier; and it will treat each incident as independent, not modeling multi-incident cascade correlation.

HELIOS is not claimed as a general-purpose RCA solution; it is claimed as an experimental artifact engineered to support falsifiable hypothesis testing on six named contributions (C1 through C6) spanning four contribution types in the Gregor and Hevner (Gregor & Hevner, 2013) taxonomy: an instantiation (HELIOS itself as an integrated artifact), a methodological-discipline contribution (C1 — runtime-enforced evaluation rigor), constructs and models (C2 — UEG-C with cognitive nodes; aspects of C4), and methods (C3′, C4, C5, C6). Confirmatory or exploratory status for each contribution is determined by the corpus feasibility documented in Section 3.5 and Section 3.6.8. In the planned evaluation, HELIOS's absolute performance against baselines is of interest but is not the primary inferential target; the primary inferential target is the pattern of ablation effects, which will license the claim that each contribution individually provides the hypothesized mechanism under

the conditions evaluable on the available corpus. The ablation-first framing distinguishes HELIOS from comparator systems engineered for production deployment first and evaluated for research contribution second, and it will justify the depth of internal instrumentation (Variant Control Layer, snapshot hash registry, reconciliation ledger, exclusion ledger) that is architectural in HELIOS but absent from the comparator corpus.

1.7.2 Architectural Design Principles

Five design principles will govern the HELIOS architecture and, through it, the methodology of the study; the principles are intended as transferable design knowledge contributions in the (Gregor & Hevner, 2013) sense, generalizable beyond HELIOS to AIOps research-method discipline more broadly.

Pipeline independence. The three reasoning modalities — D-pipe, G-pipe, L-pipe — will be peer pipelines that consume the same evidence snapshot but share no runtime state; any subset can be disabled by feature flag without affecting the operation of the others. This is the minimum structural commitment that admits causal identifiability of each paradigm's contribution to RCA performance, subject to the caveat that the absence of cross-pipeline runtime state coupling does not preclude shared selection biases inherited from the common UEG-C input.

Ablation-first design. Every mechanism claimed as a contribution must be ablatable via a feature flag, and every flag must flip exactly one well-defined code path without composing with another flag as a side effect. The twelve feature flags specified in the architecture (‘l2c_llm‘, ‘p4_cognitive‘, ‘mahc_hierarchical‘, ‘cbr_routing‘, ‘l2b_graph‘, ‘acp_stateful‘,

'reconcile_terminal', 'fgsv_ks_gate', 'iol_self_telemetry', 'coe_narrative', 'his_tier', 'sampling_regime_tag') will produce eight confirmatory variants through pre-registered combinations and will be subject to a hidden-coupling audit verifying that each flag toggles disjoint code paths. The audit has been conducted at design freeze: it identified one hidden coupling in the original 'p4_cognitive' specification (silent removal of cognitive nodes from G-pipe's input), which has been resolved by restricting the flag to L-pipe scope, and confirmed that the remaining eleven flags toggle disjoint code paths.

Bounded reproducibility under explicitly disclosed determinism conditions. D-pipe and G-pipe will be reproducible to the level of bitwise output equality under the binding hardware configuration (NVIDIA A100 80GB SXM4; CUDA 12.1; PyTorch deterministic-algorithm flags enabled via 'torch.use_deterministic_algorithms(True)' and 'CUBLAS_WORKSPACE_CONFIG=:4096:8'; cuDNN deterministic mode enabled). L-pipe reproducibility will be specified at two protocol levels rather than as bitwise. Under Protocol A (self-hosted open-weight model), the binding model is Llama-3.1-70B-Instruct served via vLLM 0.6.x under greedy decoding (temperature = 0, top-p = 1, top-k = 1) with 'enforce_eager=True', 'disable_custom_all_reduce=True', and single-replica serving; under these conditions the emitted token sequence will be reproducible across runs, but token-level logits will exhibit small floating-point variance ε due to non-deterministic CUDA attention-kernel selection and cuBLAS workspace allocation. ε will be measured during the Chapter Four pilot pass on a 200-prompt probe set and reported in Chapter Four with its standard error and measurement date; the measurement is conducted prior to chapter submission so that ε is a delivered empirical result at viva. Under Protocol B (commercial API access), neither token-sequence nor logit-level determinism is

achievable through documented means; Protocol B will be reported as a sensitivity analysis with inter-run variance characterized by the same probe-set methodology. Container digest, weights SHA-256, and tensor-parallel-degree configuration will be deposited at OSF and verified at runtime by the Variant Control Layer.

Separation of diagnosis from action. The decision layer will emit a ranked root-cause list and a proposed remediation plan; the action-and-learning layer will be the sole execution boundary. This separation allows diagnostic quality to be evaluated independently of action safety, which is methodologically necessary because the RCA accuracy hypotheses target the decision layer alone, whereas the safety and validation hypotheses target the action layer.

Failure-as-covariate discipline. Every failure mode — sampling downgrade, pipeline unavailability, consensus non-convergence, twin staleness, privacy-budget exhaustion, adversarial input, hardware contention — will be converted into a first-class observable covariate that stratifies the evaluation, not into a silent fallback. The principle is enforced by the metric integrity gate at the Variant Control Layer, which will refuse to accept a run whose covariates are not fully populated. The proportion of incidents with complete covariate records is itself a reported metric with a target of 100 percent.

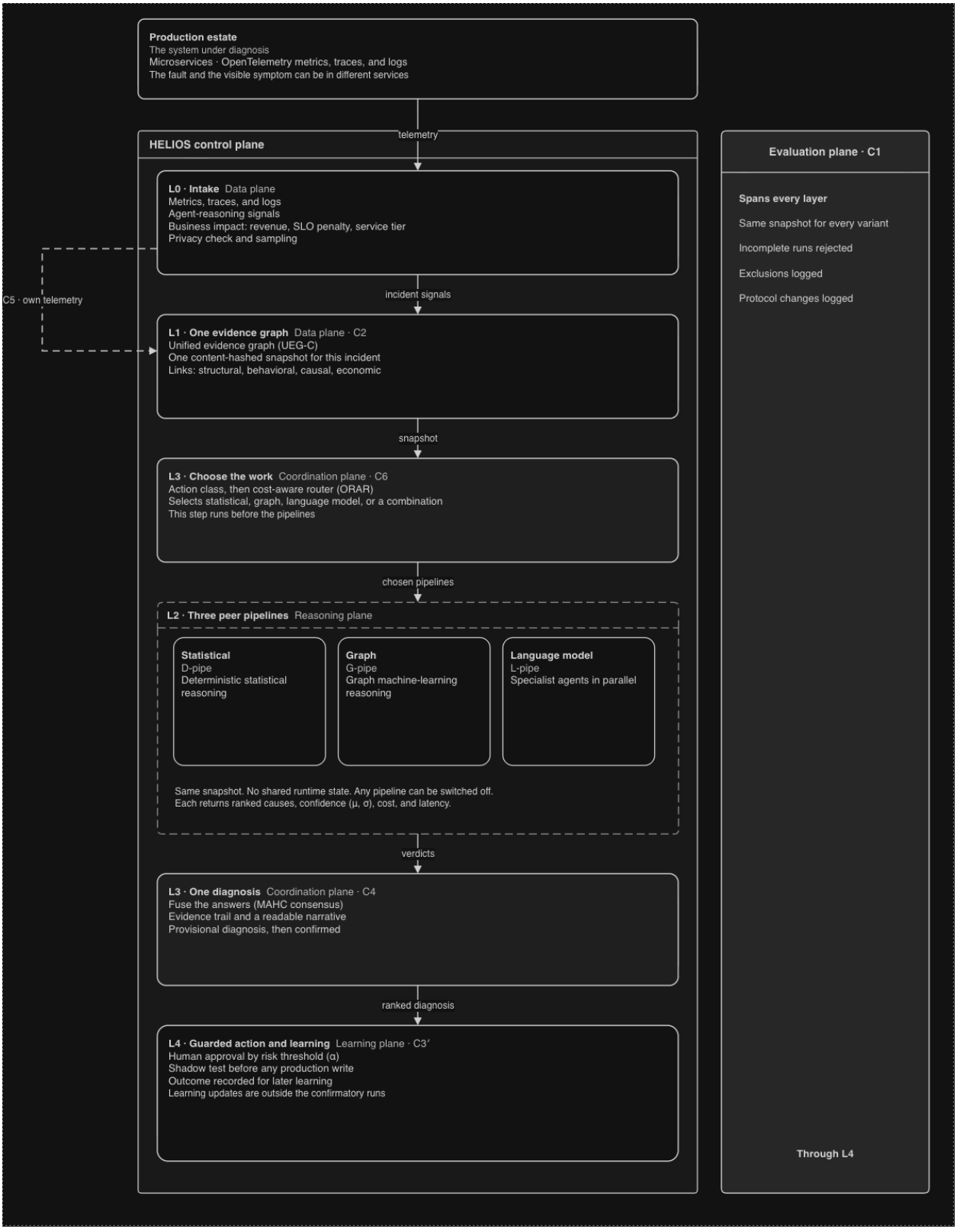
1.7.3 System Architecture Specification

The following describes the HELIOS architecture as specified at design freeze. Evaluation activities described in subsequent sections will exercise this architecture under the pre-registered

protocol; present-tense statements in this section therefore describe the design specification, not currently-running infrastructure.

HELIOS will be organized as a five-layer execution hierarchy (L0–L4) orthogonal to five governance planes (Data, Reasoning, Coordination, Learning, Evaluation), with the Evaluation plane spanning all layers. Layer L0 will host telemetry ingress (OTEL collectors, the Privacy Gateway, the Adversarial Robustness Monitor, Tiered Adaptive Sampling, and the Internal Observability Loop, which writes HELIOS's own control-plane telemetry into the L1 graph under a tenant=self partition). Layer L1 will construct and maintain the UEG-C hypergraph, with the K-hop Pruner extracting incident-local subgraphs through personalized PageRank with Pearson correlation pruning. Layer L2 will host the three peer reasoning pipelines, each consuming the same content-hashed UEG-C snapshot and emitting a typed verdict of the form (ranked_RCA_list, μ, σ, cost, latency). Layer L3 will execute coordination: the Action-Class Predictor will emit a stateful-action feature block that the ORAR router consumes; the ORAR router will select the pipeline subset subject to cost minimization under accuracy and latency constraints; the MAHC consensus will aggregate the pipeline verdicts under hierarchical priors with (μ, σ) vote representation; the Chain-of-Evidence ledger will record the decision audit trail; and the Narrative Engine will render the CoE into human-readable form. Layer L4 will execute the action-and-learning cycle: the HITL gate will stratify by dynamic autonomic threshold α; the FGSV shadow validator will execute the proposed remediation in a Kolmogorov–Smirnov-gated digital twin; the SEAN library will provide versioned remediation templates; and the reconciliation ledger will receive provisional and terminal rewards and apply compensating updates to the ORAR parameters during exploratory subsets only.

Figure 1
system-architecture.png



Each component is specified by its inputs, outputs, and ablation controllability. The OTEL collectors and Privacy Gateway are supporting infrastructure rather than ablatable contributions. The UEG-C graph store (C2) is ablated at the L-pipe-visibility level by ‘p4_cognitive‘ and at the structural level by ‘ueg_structural‘. The three L2 pipelines are ablated individually by ‘l2c_llm‘, ‘l2b_graph‘, and the implicit D-pipe-only configuration. The ACP is ablated by ‘acp_stateful‘; the CBR by ‘cbr_routing=uniform‘ (with auxiliary ablations for ‘reconcile_terminal‘ and ‘his_tier‘); MAHC (C4) by ‘mahc_hierarchical‘; FGSV (C3′) by ‘fgsv_ks_gate‘; and IOL (C5) by ‘iol_self_telemetry‘.

The L-pipe multi-agent system warrants specific attention because its architectural differentiation from prior multi-agent RCA systems is central to the contributions claimed in this dissertation. The L-pipe is organized around a Reflective Orchestrator that loads matching Standard Operating Procedures from the Knowledge Cortex, evaluates the Adaptive Specialization Gate decision function, and dispatches worker agents with Persona-Profile-bound tool permissions. The worker mesh comprises specialist agents for metric-domain anomalies, trace-domain propagation analysis, log-domain evidential reasoning, cognitive-telemetry sentinel monitoring, value-impact assessment, ethics auditing, and reward computation. This organization differs from mABC (W. Zhang et al., 2024), in which agents vote with scalar contribution × expertise weights, by emitting (μ, σ) tuples that propagate uncertainty into the MAHC consensus rule; from TAMO (X. Zhang et al., 2025), in which an expert agent synthesizes tool outputs into a single narrative, by treating agents as peer voters rather than as a single synthesizer; and from Flow-of-Action (Pei et al., 2025), in which a MainAgent orchestrates sub-agents sequentially, by operating specialists in parallel with independent evidence streams entering the MAHC consensus.

1.7.4 Data Flow and Control Flow

Data flow and control flow are deliberately separated to facilitate causal identifiability at each stage of the incident lifecycle. The data flow will commence with signal emergence at L0, where five-pillar telemetry is continuously collected at the Tier A baseline rate of 0.1 percent via eBPF kernel probes. Anomaly detection at the D-pipe will trigger sampling escalation to Tier C within the affected K-hop neighborhood, with backpressure-aware downgrade to Tier B on resource pressure (Section 3.7). The K-hop Pruner will extract a subgraph of 50 to 200 nodes with four typed edge classes, content-hash the snapshot, and publish the snapshot identity to the registry for downstream consumers.

Control flow, which determines which pipelines execute on the snapshot, will be routed through the ACP and the ORAR CBR. The ACP will read the snapshot and the incident signature and emit a stateful-action feature block including schema-change probability, data-mutation probability, config-change probability, a log-scale write-ahead-log (WAL) size estimate, and a blast-radius tier classification. The ORAR router will combine incident features, per-pipeline accuracy priors, and the ACP output into a constrained-bandit decision over the action set {D, G, L, D+G, D+L, G+L, Full}, minimizing expected cost subject to an accuracy-probability constraint at target HR@3 and a latency constraint. During the confirmatory protocol, the LinUCB per-arm parameters are frozen at the pre-warmed state (Section 3.3.3 and Section 3.6.5), so the routing decision is a deterministic function of incident features and frozen parameters, and the sequence in which incidents are processed does not influence subsequent routing. If the constraint set is empty, the router will fall back to HELIOS-Full with an explicit `constraint-empty-fallback` covariate logged to the CoE.

Once the pipeline subset is dispatched, the pipelines will execute concurrently and each will produce a typed verdict. The MAHC consensus at L3 will aggregate the verdicts under hierarchical priors (fault-type → service-family → global, with backoff triggered by minimum-sample thresholds) and emit a Fast-Path PROVISIONAL verdict within 30 seconds and a CONFIRMED verdict upon full consensus convergence; non-convergence within 90 seconds will trigger elevated-uncertainty PROVISIONAL publication plus HITL escalation, and the non-convergence count will be a reported metric per cell rather than an evaluation failure.

Action flow will proceed through the FGSV at L4, which will test the proposed remediation in the shadow twin only when the Kolmogorov–Smirnov two-sample test on pillar-matched samples confirms that twin telemetry is statistically close to production. A KS-gate failure will yield a STALE verdict, which is itself a first-class covariate; incidents with `shadow_suspended=true` will be reported separately and no action-safety claims will be made for them. Actions that pass FGSV and clear the HITL gate during evaluation runs will proceed to actuation; the actuation result will feed the provisional-reward computation approximately one to five minutes later, and post-mortem labels arriving hours to days later will drive the terminal-reward computation, the compensating ORAR parameter update, the RRE logging, and the weekly bias-audit recalibration. These on-line update operations will be performed only on the disjoint exploratory delayed-label subset described in Section 3.3.5 and will not modify the frozen LinUCB state used in the confirmatory protocol.

1.7.5 Formal Algorithms

Four formal algorithms are named as core intellectual content of HELIOS, with one supporting algorithm (the K-hop Pruner). Each is summarized below at a level sufficient for methodological grounding; full pseudocode is documented in the architecture chapter (Chapter Four). Al-

gorithm specifications use present tense in the sense of computer-science algorithm description (algorithms describe a temporal computations); evaluation activities that exercise the algorithms during the confirmatory protocol are stated in future tense.

Algorithm 1. ORAR Router. Solves a cost-minimizing constrained bandit problem at each incident. Given incident features x (augmented with the ACP stateful-action feature block), per-arm LinUCB parameters (A_a, b_a) for a □ {D, G, L, D+G, D+L, G+L, Full}, and thresholds (τ_acc, τ_lat), the router will compute the expected cost μ_cost(a) with an upper-confidence-bonus correction, evaluate a calibrated accuracy probability p̂_acc(a) and a latency estimate ℓ̂(a), construct the feasible action set, and select the feasible arm minimizing the cost-minus-bonus objective. After actuation, a provisional reward at T + Δ□ ≈ 1–5 min will be computed from the HITL-acceptance indicator, verdict self-consistency, and realized CpR; per-arm parameters will be tentatively updated only on the disjoint exploratory subsets used for ORAR learning analyses. At T + Δ□ (hours to days later), a terminal-reward correction Δr will be computed from post-mortem-label correctness; an idempotent compensating update will be applied; and the RRE per incident will be logged.

The static-policy commitment in Algorithm 1 binds in the confirmatory protocol: for the 16,000-run confirmatory ablation protocol and the 174-fault baseline-comparison protocol, the LinUCB parameters (A_a, b_a) will be frozen at the pre-warmed state established prior to OSF freeze (Section 3.3.3), and no on-line updates will be applied. Under the static policy, the k-th incident's routing decision is a deterministic function of incident features and frozen parameters, preserving the within-pair independence assumption of the paired-comparison design. The pre-warming pass operates LinUCB under thawed updates with the same exploration coefficient LinUCB used in the confirmatory protocol on the 1,200-incident synthetic corpus in pre-registered seed order; the ACP

is trained by minimizing cross-entropy against synthetic ground-truth remediation labels with early stopping on a 20 percent held-out validation subset. The pseudocode, hyperparameters, and seed sequence are deposited at OSF with content hashes recorded at freeze.

Algorithm 2. Action-Class Predictor. A small supervised classifier that pre-predicts the probable remediation class of an incident before pipeline dispatch. Given the L1 snapshot and the incident signature, it will extract structural features (node count, edge-class counts, anomaly-seed type, recent-deployment overlap, fault-signature hash, pillar-availability flags) and produce softmax probabilities over {schema change, data mutation, config change}, an MLE estimate of log expected WAL-replay size, and a blast-radius tier classification. In production deployment, the ACP would be retrained weekly on accumulated (incident, realized remediation) pairs; for the confirmatory protocol, the ACP weights will be frozen at the pre-warmed state established prior to OSF freeze (Section 3.3.3 and Algorithm B.1 in Appendix B), consistent with the LinUCB parameter freezing under the static-policy commitment. The ACP is ablated by HELIOS-noACP, which replaces it with a uniform latency prior.

Algorithm 3. Multi-Agent Hybrid Consensus. Aggregates the verdicts of active pipelines into a ranked root-cause list with calibrated uncertainty. For each pipeline p in the active set, MAHC will select a prior weight from the hierarchical prior (fault-type priors if sufficient data, else service-family priors, else global priors) and apply an uncertainty adjustment that reduces the effective weight as σ_p grows. Candidate causes from the union of pipeline ranked lists will receive a weighted Borda aggregation score; their variance will be computed as a weighted sum of squared σ_p. An entropy regularizer will test the normalized score distribution: if entropy is below

a pre-registered floor λ, MAHC will emit PROVISIONAL plus a HITL request rather than auto-executing. The mechanism is ablated by HELIOS-noConsensus, isolating the effect of uncertainty-aware hierarchical consensus on HR@3 and on the consensus-confusion rate.

Algorithm 4. Fidelity-Gated Shadow Validation. Validates a proposed remediation in the shadow twin only when twin-to-production distributional distance, measured by the two-sample Kolmogorov–Smirnov test on pillar-matched samples, falls below a pre-registered threshold $\alpha_{ks} = 0.05$ (one-sided rejection; with companion Type II bound $\beta_{ks} = 0.20$ for a Cohen's $d = 0.3$ target on the KS-statistic effect). When the KS gate passes, the remediation will be executed in the twin under the Action-Scoped Replay Engine (ASRE) mode (structural, WAL-replay, or probabilistic sketch) selected by the remediation's action class; post-action twin telemetry will be captured, and blast-radius will be computed against the pre-action baseline. Blast-radius exceedance triggers a FAIL verdict; KS-gate failure triggers a STALE verdict tracked as a first-class covariate.

Algorithm 5. K-hop Pruner (supporting). Given the UEG-C graph, seed nodes, a hop limit K, a Pearson threshold (pre-registered at $r = 0.7$), and an edge budget of 10,000, it expands the seed set by K-hop traversal over structural and behavioral edges, retains top-n nodes by personalized PageRank restart probability, drops causal and behavioral edges below the correlation threshold, and content-hashes the resulting subgraph for snapshot registry storage. The snapshot hash is the content-addressable identity under which between-variant comparisons are valid.

1.7.6 Observability Planes and Layers

HELIOS maintains two orthogonal structural decompositions: the five-layer execution hierarchy (L0–L4; Section 3.6.3) and a five-plane governance decomposition (Data, Reasoning, Coordination, Learning, Evaluation). Each component has a unique layer and a unique plane; the Evaluation plane deliberately spans every layer because it observes and gates every layer. Within the Evaluation plane, Class 1 components affect runtime behavior and together constitute Contribution C1 (Section 3.6.8): the Variant Control Layer (binary-level variant identity), the snapshot hash registry (input identity across variants), the metric integrity gate (runtime rejection of incomplete cells), the exclusion ledger (cryptographically-signed append-only missingness record), the deviation log (cryptographically-signed append-only protocol-change record), and the reconciliation ledger (provisional-and-terminal reward correspondence). Class 2 components observe but do not modify runtime behavior and are experimental infrastructure operating under OSF pre-registration discipline. The IOL self-telemetry path deliberately spans all layers under the tenant=self partition, enabling the methodological claim that HELIOS's own degradation will be observable through the same diagnostic apparatus that HELIOS applies to production. The empirical validation of C5 rests on synthetic injected control-plane degradation and is reported as exploratory rather than confirmatory evidence (Section 3.6.8).

1.7.7 Ablation-Aware Architecture

The ablation-aware architecture is the methodological core of the artifact and therefore of the research design. The twelve feature flags documented in Section 3.6.2 will produce eight confirmatory variants consistent with the OSF pre-registration (Table 2) and seven exploratory variants

documented below. None of the flags composes with another as a side effect, subject to the hidden-coupling audit reported in Section 3.6.2.

Table 5
Feature-Flag Matrix for the Eight Confirmatory Variants

Variant	L2C (LLM)	P4 (Cogn.)	MAHC	CBR	L2B (Graph)	ACP	Reconcile	UEG-C Struc-tural	Primary Test
HELIOS-Full	ON	ON	ON	ON	ON	ON	ON	ON	Reference
HELIOS-noLLM	OFF	ON	ON (D+G)	ON {D, G}	ON	ON	ON	ON	A-H1 — LLM neces-sity
HELIOS-LLM-only	ON	ON	passthru	passthru	OFF	OFF	passthru	ON	A-H2 — LLM suffi-ciency

Variant	L2C (LLM)	P4 (Cogn.)	MAHC	CBR	L2B (Graph)	ACP	Reconcile	UEG-C Struc-tural	Primary Test
HELIOS-D	OFF	ON	passthru	passthru	OFF	OFF	passthru	ON	A-H3 — Statistical floor
HELIOS-noConsensus	ON	ON	OFF (vote-only)	ON	ON	ON	ON	ON	A-H4 — Uncertainty consensus value
HELIOS-noRouter	ON	ON	ON	uniform	ON	ON	ON	ON	A-H5 — C6 routing value

Variant	L2C (LLM)	P4 (Cogn.)	MAHC	CBR	L2B (Graph)	ACP	Reconcile	UEG-C Struc-tural	Primary Test
HELIOS-G	OFF	ON (graph enc.)	passthru	passthru	ON	OFF	passthru	ON	A-H6 — Graph-only on narrative-heavy
HELIOS-noLLM (factual hallu-cination)	OFF	ON	ON (D+G)	ON	ON	ON	ON	ON	A-H7 — Hal-lucination zero with-out LLM

Variant	L2C (LLM)	P4 (Cogn.)	MAHC	CBR	L2B (Graph)	ACP	Reconcile	UEG-C Struc-tural	Primary Test
HELIOS-noStructural	ON	ON	ON	ON	ON	ON	ON	OFF (com-posi-tion + train-ing dis-abled)	A-H8 — C2 struc-tural and integra-tion value

Note. "passthru" denotes that the named mechanism is inactive in the variant because its upstream dependencies are absent. The p4_cognitive flag is scoped to L-pipe visibility per the hidden-coupling audit (Section 3.6.2).

When set to OFF in HELIOS-noStructural, the ueg_structural flag replaces the four typed edge classes with a single untyped edge class, disables the subgraph-aware training discipline (each of D-pipe and G-pipe is trained on the full UEG-C rather than on the K-hop pruned subgraph consumed at inference time), and demotes Agent-Reasoning-Step nodes to flat textual descriptions presented to L-pipe agents. HELIOS-noStructural retains all other HELIOS-Full mechanisms and is therefore the closest minimal ablation isolating C2's structural and integration contribution. A-H7 is tested on the same HELIOS-noLLM variant as A-H1 because the hallucination-zero hypothesis

logically follows from removing the LLM; the two hypotheses use the same variant configuration but evaluate distinct dependent variables (HR@3 for A-H1, H_fact and H_struct for A-H7) and are therefore testable on the same paired observations without methodological conflict. A-H8 is a bundle-level confirmatory test: HELIOS-noStructural simultaneously ablates the four typed-edge composition rules, the subgraph-aware training discipline, and the structured Agent-Reasoning-Step node specification, and a single confirmatory hypothesis cannot identify which of these three sub-mechanisms carries any observed effect. Within-bundle attribution is reported descriptively under exploratory hypothesis E-H10 (Section 3.8.2) and is not part of the confirmatory family. Across all eight confirmatory variants, the ORAR LinUCB parameters and the ACP weights are frozen at the pre-warmed state (Sections 3.3.3 and 3.6.5).

Seven exploratory variants are also defined, each addressing a hypothesis demoted to exploratory status because the confirmatory corpus does not support its evaluation under the pre-registration discipline: HELIOS-noCogni (E-H1, cognitive-agentic HR@3 on the n = 12 Tier 4 subset, below statistical minimum), HELIOS-noACP (E-H8, latency-constraint behavior), HELIOS-noReconcile (E-H5, RRE on the delayed-label subset, requiring synthetic delayed-label injection), HELIOS-HIS3 (E-H9, hardware-isolation contention bias, an engineering-validity probe), HELIOS-noFGSV (E-H4, false-accept rate under synthetic fidelity-break perturbation), HELIOS-noIOL (E-H6, degradation-detection latency under injected control-plane degradation), and HELIOS-noExpl (E-H7, RQ4 user-study within-subjects narrative outcome). Exploratory variants will be reported with point estimates and 95 percent Bias-corrected and accelerated (BCa) bootstrap confidence intervals, clearly labeled and visually separated from the confirmatory cells, and will not enter the Holm–Bonferroni correction family. The reasons for exploratory placement are corpus-feasibility constraints documented in Section 3.6.8: RRE requires delayed-label

scenarios absent from the benchmarks as shipped (addressed via synthetic delayed-label injection on a 25 percent subset); FGSV and IOL require synthetic perturbation protocols (fidelity breaks and injected control-plane degradation) that warrant separate future pre-registration; HIS3 is an engineering-validity measurement; and noExpl belongs to the Research Question 4 user study.

1.7.8 Embedded Novelty

The artifact's novelty is embedded in six named contributions C1 through C6, ablation-tested where corpus feasibility permits and exploratorily validated where it does not. The present section summarizes the novelty attribution in a disciplined format: what is new precisely, how it differs from its most proximate comparator, how it is ablated or evidenced, and the evidential status (confirmatory or exploratory) of its empirical evaluation under the present corpus. Contribution-type classification follows (Gregor & Hevner, 2013): C1 is a methodological-discipline contribution; C2 is a construct-and-model contribution; C3′ is an evaluation-infrastructure-discipline contribution; C4 is a method contribution; C5 is an architectural-pattern contribution; and C6 is a method contribution with a cross-field-transfer profile from the delayed-feedback bandit literature.

C1 — Runtime-Enforced DSR Evaluation Rigor. Novel in the architectural enforcement of pre-registration discipline through the integrated Class 1 component apparatus: the Variant Control Layer enforces variant identity at the binary level via content-hashed feature-flag manifest reading at process start; the snapshot hash registry enforces input identity across variants via content-addressed UEG-C snapshots consumed by every active pipeline; the metric integrity gate enforces measurement completeness via runtime rejection of incomplete cells; the exclusion ledger enforces transparent missingness via cryptographically-signed append-only logging of ev-

ery excluded run; the deviation log enforces protocol-change transparency via cryptographically-signed append-only logging of every departure from the pre-registered plan; and the reconciliation ledger enforces deferred-reward correctness via provisional-and-terminal correspondence audit. Not novel in feature-flagging, content-hashing, ledger architectures, or pre-registration as concepts taken individually; the novelty is the integration as a research-method discipline that transforms DSR evaluation rigor from a reporting convention into a runtime invariant of the artifact. The most proximate comparator is the comparator AIOps evaluation literature (CHASE, mABC, FrugalGPT, TAMO, Flow-of-Action), in which ablations are reported in prose without binary-level variant identity, snapshot identity is asserted but not verified, missingness is silently imputed, and pre-registration is absent. The C1 contribution is methodological rather than algorithmic: it does not improve diagnostic accuracy by any pipeline mechanism, but it makes the falsifiability of every other contribution operationally enforceable. C1 is evaluated descriptively rather than via a confirmatory ablation hypothesis, since the contribution is the runtime-enforcement apparatus itself rather than a flag-toggleable mechanism; the enforcement is observed by the verification-rate, completion-rate, and ledger-entry-count metrics specified in Table 1 and reported as descriptive evidence in subsequent chapter. C1 is also a transferable-knowledge contribution in the Gregor and Hevner (Gregor & Hevner, 2013) sense, generalizable beyond HELIOS to AIOps research-method practice and arguably to DSR research-method practice more broadly.

C2 — UEG-C with cognitive nodes. Novel in the introduction of Agent-Reasoning-Step nodes and Transaction-or-KPI nodes into the evidence graph, the four typed edge classes (structural, behavioral, causal, economic) with formal composition rules, and subgraph-aware training under identical pruning at training and inference time. Not novel in the heterogeneous

graph-neural-network backbone (adopted from CHASE; (Zhao et al., 2025), in personalized PageRank (adopted from MicroRCA; (L. Wu et al., 2020), or in hypergraph learning. The novelty is architectural-plus-system-integration. The mechanism is decomposed into two distinguishable claims with separate evidential pathways: a structural-and-integration claim (typed-edge composition rules, Agent-Reasoning-Step node specification, subgraph-aware training discipline), tested confirmatorily through HELIOS-noStructural under hypothesis A-H8 (Table 2); and a cognitive-agentic-accuracy claim (improvement on agentic-AI fault classes specifically), reclassified as exploratory empirical validation under E-H1, because the agentic-AI fault corpus consists of the 12 researcher-constructed Tier 4 instances — below the 73-fault statistical minimum, constructed through NIKA-style instrumentation specifically to exercise cognitive-node reasoning, and layered onto an AIOpsLab substrate that further bounds E-H1's exploratory evidence to "AIOpsLab plus cognitive-agentic instrumentation" rather than to cognitive-agentic faults in general. A-H8 is positioned as a bundle-level confirmatory test of the structural-and-integration contribution as an integrated whole. The variant ablates three sub-mechanisms simultaneously — typed-edge composition, subgraph-aware training, and structured Agent-Reasoning-Step nodes — and the chapter accepts that within-bundle attribution among these three sub-mechanisms is exploratory rather than confirmatory; descriptive within-bundle attribution is reported under E-H10 in the exploratory family.

C3′ — FGSV. An evaluation-infrastructure-discipline contribution with exploratory empirical validation. Novel in (a) the KS-gated shadow-suspension protocol with pre-registered $\alpha_{ks} = 0.05$ and $\beta_{ks} = 0.20$ for a Cohen's $d = 0.3$ target on the KS-statistic effect, (b) the STALE verdict as a first-class covariate enabling stratified analysis of shadow-validation availability rather

than a silent fallback, and (c) the bundling with action-class routing across the three ASRE modes (structural, WAL-replay, probabilistic-sketch). Not novel in traffic mirroring (an Istio deployment pattern), in the KS test, in shadow-testing as a concept, or in any individual ASRE mode. The contribution is evaluation-infrastructure discipline rather than algorithmic invention: FGSV converts shadow-testing from an engineering pattern into a falsifiable research instrument with pre-registered Type I and Type II bounds and a structurally tracked stale-fidelity covariate. The mechanism will be exercised by HELIOS-noFGSV under a synthetic fidelity-break perturbation protocol that does not arise spontaneously from the public benchmarks. Empirical evidence on the false-accept rate will be reported as exploratory (E-H4) and motivates a separate future pre-registration with a pre-specified effect size; the present chapter does not claim confirmatory empirical validation of C3′ on the available corpus.

C4 — MAHC. Novel in the (μ, σ) vote representation, the three-level hierarchical priors with backoff, and the entropy regularizer against overconfident unanimity. Not novel in weighted voting (adapted from mABC; (W. Zhang et al., 2024) or in Borda aggregation. The novelty is algorithmic, the primary comparator is mABC, and the mechanism will be ablated confirmatorily by HELIOS-noConsensus on HR@3 (ambiguous-incident stratum) and on the consensus-confusion rate under hypothesis A-H4. A deterministic synthetic counterexample — three-agent agreement at $\sigma = 0.1$ versus three-agent agreement at $\sigma = 0.8$, yielding identical scalar votes but distinguishable (μ, σ) representations — will be reported alongside the confirmatory inference as a conceptual illustration of the representation gap, but is not positioned as confirmatory evidence; the gap exists conceptually, but whether it matters empirically on the corpus is the question A-H4 and A-H7 are designed to answer.

C5 — IOL. An architectural contribution with exploratory empirical validation. Novel in writing self-telemetry to the same L1 graph under tenant=self and in consuming that telemetry through the same L2 pipelines that perform production diagnosis, without a separate monitoring stack. Not novel in self-monitoring as a concept. The most proximate comparators are concrete control-plane self-monitoring patterns rather than AIOps RCA systems: Prometheus monitoring its own scrape pipeline through self-instrumentation, OpenTelemetry self-instrumentation of its collector and exporter components, and the meta-monitoring features of commercial observability platforms such as Datadog and New Relic. Each of these patterns produces self-telemetry as a separate signal stream consumed by a separate monitoring layer; what differentiates IOL is not self-monitoring as a concept but the integrated-graph treatment in which self-telemetry is written under tenant=self into the same UEG-C consumed by the production-diagnostic pipeline. The integrated-graph treatment supports cross-tenant causal-path queries that flat self-monitoring patterns cannot express because their self-telemetry is structurally segregated from production telemetry. Three illustrative query classes are: (i) a path query from a production-tenant anomaly node through a behavioral edge to a control-plane-tenant degradation node within a single graph operation (e.g., correlating an L-pipe latency anomaly with an LLM Gateway saturation event); (ii) a typed-edge composition query joining causal edges from production-tenant anomalies with economic edges from business-tenant SLO breaches and structural edges from control-plane-tenant component dependencies (e.g., quantifying the SLO-breach contribution attributable to control-plane reasoning failures rather than data-plane infrastructure failures); (iii) a cross-tenant entropy query measuring whether a control-plane reasoning failure (e.g., MAHC consensus non-convergence) propagates to multiple production tenants simultaneously, distinguishing systemic from tenant-specific control-plane degradation. The C5 contribution is therefore a comparative claim with an articulated

mechanism rather than a uniqueness-by-absence claim, even though the empirical validation of the mechanism remains exploratory pending injected control-plane degradation under E-H6.

C6 — ORAR. A method contribution with a cross-field-transfer profile from the delayed-feedback bandit literature (Joulani et al., 2013) to AIOps RCA, where the source literature does not include the integrated extensions claimed here. Novel in (a) the two-stage provisional-and-terminal reward with reconciliation ledger, transferred from the delayed-feedback tradition but absent from the RCA literature; (b) the pre-routing Action-Class Predictor for latency-aware arm selection, with no prior AIOps system precedent; and (c) hardware-isolation-tier (HIS) tagging of every latency and cost measurement, a measurement-validity contribution with no precedent in the comparator set. Not novel in LinUCB (Li et al., 2010), in cost-aware cascading (FrugalGPT (L. Chen et al., 2023)), or in the contextual-bandit-routing concept. The primary comparator is FrugalGPT. The mechanism will be ablated confirmatorily by HELIOS-noRouter (A-H5) under the static-policy commitment, and exploratorily by HELIOS-noACP, HELIOS-noReconcile, and HELIOS-HIS3 (E-H8, E-H5, E-H9). The static-policy commitment narrows the confirmatory inference for C6 to the cost-aware static routing mechanism; the on-line learning behavior is reserved for the exploratory subset, and a confirmatory test of the on-line dynamics would require a future pre-registration that resolves the paired-inference confound of cross-incident learning.

The contribution-level evidential status under the present empirical evaluation is summarized as follows: C1 — methodological-discipline contribution evidenced descriptively through verification-rate, completion-rate, and ledger-entry-count metrics (Section 3.3.6 Table 1); C2 — confirmatory for the structural-and-integration claim through A-H8, with concurrent confirmatory evidence on hallucination rates through A-H7; exploratory for the cognitive-agentic-accuracy

claim through E-H1; exploratory for within-bundle attribution through E-H10; C3′ — evaluation-infrastructure-discipline contribution with exploratory empirical validation through E-H4; C4 — confirmatory through A-H4 (HR@3 on ambiguous incidents); C5 — architectural-pattern contribution with exploratory empirical validation through E-H6; C6 — confirmatory for static cost-aware routing through A-H5, exploratory for on-line learning dynamics through E-H5. This profile reflects explicit alignment between corpus feasibility (Section 3.5) and contribution claims, and is mapped to the hypothesis structure in Section 3.8.2 and to the limitations in Section 3.11.

1.8 Data Collection Procedures

Data collection will operate at two distinct temporal and architectural scales: continuous telemetry ingestion during each incident run, and per-run experimental-record aggregation across the 16,000-run ablation protocol and the 174-fault baseline-comparison corpus. The two scales are coupled by the run orchestrator, which will manage the dispatch schedule, record completion state per variant–benchmark–metric cell, and block aggregation until a cell is either completed or formally excluded under the inclusion criteria specified in the pre-registration.

Telemetry ingestion will proceed continuously across all five pillars via OTEL-compliant protocols (OTLP gRPC and HTTP). The Privacy Gateway will apply pattern-based personally identifiable information (PII) detection and named-entity-recognition-based redaction at the Data-Plane-to-Control-Plane boundary before any signal is forwarded to Plane 1 aggregators; redacted tokens will be replaced with deterministic placeholders that preserve grammatical structure for downstream LLM parsing. Per-tenant differential-privacy budgets ($\varepsilon = 1.0$ per tenant, $\delta = 10^{\square\square}$, with budget renewal at 24-hour rolling windows) will cap total information leakage from aggregated Knowledge Cortex queries; the (ε, δ) values are pre-registered at OSF freeze and any devia-

tion recorded in the deviation log. The Adversarial Robustness Monitor will inspect all ingress in parallel for prompt-injection patterns, telemetry-poisoning signals, and anomalies in the telemetry emission process itself. Data-collection volume is governed by the three-tier adaptive sampling strategy (Tier A baseline at 0.1 percent via eBPF kernel probes; Tier B dynamic adjustment based on runtime SLO pressure; Tier C anomaly-triggered escalation to 100 percent within the affected K-hop neighborhood). Tier C will downgrade automatically to Tier B when node memory utilization exceeds 80 percent or CPU exceeds 90 percent; the downgrade will be logged as a first-class covariate (`sampling_regime=downgraded`) so that downstream analysis can stratify by sampling regime and avoid attributing sampling-induced signal loss to an algorithmic effect.

Fault dataset will be sourced from the five pre-specified benchmarks (Section 3.5) with identical telemetry access granted to all evaluated systems, and ground-truth root-cause labels will be sourced directly from each benchmark's published annotation; no annotation will be regenerated or adjudicated for HELIOS-specific criteria. The 174-fault augmentation corpus will combine these benchmark sources with the Chaos-Mesh parametric extension (Tier 2) and the NIKA-style cognitive-agentic synthetic faults (Tier 4), with SHA-256 hashes of every fault instance and reserve-pool instance published at pre-registration freeze.

Experimental-record aggregation will operate at the variant–benchmark–metric cell granularity. Each cell will accumulate 400 paired observations (40 faults × 10 seeds) for the ablation protocol or the tier-determined count for the baseline-comparison corpus. The metric integrity gate at the Variant Control Layer will verify at run completion that every metric expected for the cell has been produced, will reject runs with missing measurements rather than silently impute zeros, and will record exclusions in the cryptographically-signed, append-only exclusion ledger. The minimum completed-run threshold for cell inclusion in confirmatory inference is 320 of 400 (80

percent); cells below this threshold will be reported as "insufficient data" with the completed count and the reason, and will not enter the hypothesis tests. No hypothesis will be dropped silently.

The 80 percent threshold (320 of 400 paired observations per cell) is justified on three grounds: (i) it preserves $n \geq 320$ paired observations per cell, exceeding the per-cell statistical minimum of $n = 73$ by 4.4× (Section 3.8.3); (ii) failed runs are characterized by infrastructure-level mechanisms (out-of-memory kills, network partitions, hardware contention, container-runtime errors) rather than fault-class-correlated mechanisms, supporting a missing-completely-at-random (MCAR) assumption; (iii) the MCAR assumption will be empirically tested in Chapter Four by comparing fault-class proportions and severity distributions between included and excluded runs at $\alpha = 0.05$ (chi-square test, two-sided). If the MCAR test rejects, the affected cells will be supplemented with sensitivity analysis using inverse-probability weighting on the propensity to be excluded, with weights estimated from the exclusion-ledger covariates; this conditional sensitivity analysis is pre-registered at OSF freeze. Hallucination labeling for H_fact and H_struct will follow the same inter-rater protocol as the RQ4 user study (two independent raters blind to variant, Cohen's $\kappa \geq 0.7$, third-adjudicator re-labeling triggered below 0.7, rubrics deposited at OSF freeze); raters will be calibrated on a 50-instance training subset prior to confirmatory labeling.

1.8.1 Exploratory Human Evaluation (Research Question 4)

The automated ablation and baseline protocols answer whether HELIOS is more accurate or cheaper than its comparators. They do not say whether a person looking at the output can actually use it. Research Question 4 is about that second problem: whether the Chain-of-Evidence narrative helps someone interpret a diagnosis, compared with a raw ranked list. The human evaluation is therefore an exploratory component of the design, not a third confirmatory family. It is bound

to hypothesis E-H7 in Table 3, uses the HELIOS-noExpl variant (HELIOS-Full with narrative rendering switched off), and is reported with point estimates and interval estimates only. It does not enter Holm–Bonferroni correction, and n = 24 is not treated as a basis for claiming how SREs, academics, or managers in general would behave.

The planned sample is 24 people, mixed on purpose. Incident output is not read only by on-call engineers. Academics who work on this literature, and business or service owners who approve tooling and live with downtime cost, also have to make sense of a diagnosis. The target mix is twelve practitioners with current or recent incident-response duties, six academic participants in information systems or software engineering, and six business or operations stakeholders (service owners, product managers, or equivalent). If a cell is hard to fill, the study will run with the mix actually recruited and the composition will be reported; roles will not be reclassified after the fact to make the table look balanced.

Recruitment will go through the researcher's professional network and through open calls on practitioner and academic lists. Screening is role-specific. Practitioners need at least three years of production incident work and a working familiarity with traces or equivalent observability tooling. Academic participants need postgraduate training or a faculty appointment in a related field and enough background to read an incident write-up. Business participants need a role in which they consume incident reports or decide on operational tooling. Anyone who helped design HELIOS, or who can identify the two output conditions from prior exposure, is excluded. A fixed honorarium, set at about one hour of professional consulting time and documented in the IRB protocol, will be paid; it is meant to compensate time, not to pressure anyone to take part.

The design is within-subjects. Each person sees both output conditions on a stratified set of twelve fault scenarios taken from the confirmatory corpus: HELIOS-Full (ranked list plus Chain-

of-Evidence narrative) and HELIOS-noExpl (the same underlying diagnosis with the narrative stripped, leaving the ranked list). Order is counterbalanced with a Latin square over output type and scenario stratum so that fatigue and learning are not loaded onto one condition. Sessions are expected to last about 60–75 minutes, remote or in person, under the same task script.

For each scenario the participant is given the RCA output and a short incident stem (what broke, roughly when, which service the user saw fail). The task is not to debug the live system. It is to (1) name the service they believe is the root cause, (2) name a fault category, (3) point to at least one piece of evidence in the output that supports that reading, and (4) choose accept, escalate, or reject as if they were the human in the loop. After each item they rate trust in that output on a five-point scale. Ground truth for scoring comes from the benchmark labels already used in the confirmatory corpus; participants do not see those labels during the task.

Three measures are collected. Correct interpretation is a binary score per person × scenario: the written answer has to get the ground-truth service, the ground-truth category, and one corroborating evidence item. Two raters, blind to condition, will score against a rubric deposited at OSF; Cohen's $\kappa \geq 0.7$ is required, and a third rater is brought in below that. Time-to-decision is the clock from presentation of the output to submission. Trust is the Likert rating, read alongside the accept / escalate / reject choice, which is logged and cannot be edited later. The Likert item is easy to talk up; the logged choice is there because of that.

Analysis follows E-H7. The binding comparison for correct-interpretation rate is a paired Wilcoxon signed-rank test. Time-to-decision is tested with a paired t on the log scale if Shapiro–Wilk at $\alpha = 0.05$ does not reject, otherwise Wilcoxon. Trust is reported as median and a 95 percent bootstrap interval per condition. Role (practitioner / academic / business) is a descriptive split, not a

confirmatory factor; cells of six to twelve people are too thin for that. As an exploratory hypothesis, E-H7 also carries a 95 percent BCa interval and stays out of the Holm families.

The study is a probe of whether the narrative rendering is usable by the kinds of people who would actually see HELIOS output, under twelve constructed scenarios. It is not a population survey of SREs, not a market study, and not evidence that organizations will adopt the system. Mixing roles will add variance that the original n = 24 calculation ($d_z = 0.6$, one-sided $\alpha = 0.05$, 80 percent power, n = 22 plus 10 percent attrition) did not fully price in; that calculation is a planning number, not a license to generalize. Findings are bounded to the people who take part and to the scenarios shown. A larger, role-stratified confirmatory study would be a separate pre-registration.

The protocol, recruitment text, consent form, debrief, and analysis plan go to the Golden Gate University IRB before any data are collected. Approval status and protocol number will be written into the OSF deviation log at freeze. Ethics handling for participants is specified in Section 3.10.

Principal data-collection instruments will be aligned with the dependent variables specified in Section 3.3.6 and the evaluation metrics specified in Section 3.8.4. Per-plane OTEL span duration histograms will provide time-to-detect (TTD), time-to-triage (TTT), and per-plane inference latency at p50, p95, and p99. The MAHC consensus output module at Plane 4 will provide the ranked root-cause list against which ground-truth labels are evaluated for HR@k, mean reciprocal rank (MRR), and macro F1. The Plane 3 LLM Gateway token counter will provide per-episode token spend for CpR computation, and the Plane 7 Critic Agent post-incident computation module will aggregate token, infrastructure, storage, human-minute, and opportunity-cost components into the five-component CpR sum. The P4 cognitive-telemetry pipeline will provide schema-validation failure counts for structural hallucination rate (H_{struct}); the Narrative Engine output, parsed by a

deterministic claim-extraction parser, will provide atomic factual claims for independent-rater labeling of the factual hallucination rate (H_fact). The Monitor Agent audit output will provide ethics-compliance rates, the Plane 5 MAML-Quick execution will provide retraining overhead in CPU-hours and GPU-hours, and the multi-judge Faithfulness-Accuracy-Explanation Score (FAE)-Score ensemble (three judge models plus 10 percent human-anchored review) will provide narrative-quality scoring for the Research Question 4 user study. Each instrument is specified by its architectural location and its event-generation contract, supporting independent replication.

1.9 Data Analysis Plan

1.9.1 Research Questions and Hypotheses

The research questions that will orient the empirical evaluation are derived from the dissertation's stated objective and from the architectural commitments of the HELIOS artifact.

Research Question 1. To what extent will the co-deployment of statistical, graph-machine-learning, and large language model reasoning as peer pipelines under identical inputs improve root-cause-analysis accuracy and reduce cost per resolution compared with baseline systems that deploy a single reasoning modality or a sequential pipeline?

Research Question 2. To what extent will the integration of self-telemetry into the same diagnostic apparatus enable the detection and mitigation of reasoning failures within the AI system itself, beyond the detection of failures in the production workloads the AI system is diagnosing?

Research Question 3. To what extent will each named architectural mechanism of HELIOS — cognitive-telemetry integration (C2), fidelity-gated shadow validation (C3′), uncertainty-aware hierarchical consensus (C4), meta-observability (C5), and cost-aware adaptive routing (C6) — individually contribute to diagnostic accuracy, cost efficiency, and safety, as determined by ablation?

Research Question 4. To what extent will the Chain-of-Evidence narrative rendering improve SRE interpretation of RCA output, measured by correct-interpretation rate, time-to-decision, and self-reported trust in a within-subjects user study?

The pre-registered hypotheses are organized into three families, each corrected for family-wise error under Holm–Bonferroni with pre-registered rank ordering specified in Section 3.8.3. The ablation family (A-H1 through A-H8) tests the internal contribution of each architectural mechanism against the corresponding feature-flag-gated variant; it contains eight confirmatory hypotheses, evaluated on the eight ablation variants specified in Section 3.6.7 Table 2 (note that A-H1 and A-H7 are tested on the same HELIOS-noLLM variant but evaluate distinct dependent variables — HR@3 for A-H1 and hallucination rates for A-H7 — and are therefore independent confirmatory tests). The baseline-comparison family (B-H1 through B-H8) tests the external comparative advantage of HELIOS-Full against five priority-ranked AIOps baselines on HR@3 (B-H1 through B-H5), log-MTTR (B-H6), log-Cost (B-H7), and Avg@5 (B-H8); the baseline family is corrected separately from the ablation family because the two families answer different questions (internal mechanism contribution vs external comparative advantage) and would violate the inferential logic of a single correction family if combined.

The exploratory family (E-H1 through E-H10) does not enter Holm–Bonferroni correction and is reported with point estimates and 95 percent BCa bootstrap confidence intervals only; it in-

cludes the C2 cognitive-agentic test (E-H1, reclassified from the ablation family because the Tier 4 corpus of n = 12 is below the 73-fault statistical minimum and is constructed to exercise cognitive-node reasoning per Section 3.6.8); the demoted near-tautological hypothesis E-H2 (uniform routing increases TTT, which follows from the deselection of the latency-aware routing constraint) and the related E-H3 (latency-constraint behavior); the FGSV false-accept rate (E-H4) and IOL detection latency (E-H6), both requiring synthetic perturbation protocols absent from the public benchmarks; the RRE measurement on the delayed-label subset (E-H5); the RQ4 user-study CoE narrative outcomes (E-H7); the HELIOS-noACP latency-constraint and HELIOS-HIS3 contention exploratory variants (E-H8, E-H9); and the C2 within-bundle attribution probe (E-H10) introduced under the bundle-level framing of A-H8 (Section 3.6.8). The schema is exhaustive; every pre-registered hypothesis receives exactly one label, and the prefixed labels are binding from OSF freeze forward. C1 is evidenced descriptively rather than via a confirmatory or exploratory hypothesis (Section 3.6.8) and therefore does not appear in any of the three families.

1.9.2 Research Question–Hypothesis–Architecture Mapping

The binding mapping from research questions to hypotheses to architectural components and ablation variants is presented in Table 3. Every confirmatory hypothesis corresponds to exactly one variant–metric cell; every variant corresponds to exactly one set of feature-flag configurations as specified in Section 3.6.7; every metric has a named measurement location in the architecture as specified in Section 3.7. Exploratory hypotheses appear in the table for completeness and are clearly demarcated. The mapping is operational rather than documentary: it is enforced by the Variant Control Layer at pre-registration freeze, and the run orchestrator will block aggregation of any cell that fails to receive its full complement of paired observations under the inclusion criteria.

Table 6
Research Question to Hypothesis to Architecture to Variant to Metric to Statistical-Test Mapping

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ1	**A-H1:** If the L-pipe is removed (HELIOS-noLLM) rather than retained (HELIOS-Full), then HR@3 will be lower.	L2c L-pipe	HELIOS-noLLM	HR@3	Wilcoxon signed-rank	Ablation / Confirma-tory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ1	**A-H2:** If only the L-pipe is used (HELIOS-LLM-only) rather than the hybrid peer pipelines (HELIOS-Full), then HR@3 will be lower.	L2c L-pipe	HELIOS-LLM-only	HR@3	Wilcoxon signed-rank	Ablation / Confirma-tory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ1	**A-H3:** If only the D-pipe is used (HELIOS-D) rather than HELIOS-Full, then HR@3 will be lower.	L2a D-pipe	HELIOS-D	HR@3	Wilcoxon signed-rank	Ablation / Confirma-tory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ1	**A-H5:** If ORAR is replaced by uniform routing (HELIOS-noRouter) rather than retained, then CpR at matched HR@3 will be higher.	L3 ORAR CBR	HELIOS-noRouter	CpR at matched HR@3	Wilcoxon signed-rank	Ablation / Confirma-tory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ1	**B-H1–B-H5:** If HELIOS-Full is used in place of baseline *k* (*k* = 1…5, priority-ranked), then the paired binary HR@3 outcome will favor HELIOS-Full.	Full system	HELIOS-Full vs. 5 Baselines	HR@3 (paired binary)	McNemar's exact	Baseline / Confirma-tory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ1	**B-H6:** If HELIOS-Full is used in place of the best baseline, then log-MTTR will be lower.	Full system	HELIOS-Full vs. Best	log-MTTR	Wilcoxon signed-rank (paired t on log values reported as sensitivity analysis)	Baseline / Confirma-tory
RQ1	**B-H7:** If HELIOS-Full is used in place of the best baseline, then log-Cost (CpR) will be lower.	Full system	HELIOS-Full vs. Best	log-Cost	Wilcoxon signed-rank (paired t on log values reported as sensitivity analysis)	Baseline / Confirma-tory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ1	**B-H8:** If HELIOS-Full is used in place of the best baseline, then Avg@5 will be higher.	Full system	HELIOS-Full vs. Best	Avg@5	Wilcoxon signed-rank (paired t reported as sensitivity analysis)	Baseline / Confirma-tory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ2	**E-H6:** If IOL is disabled (HELIOS-noIOL) rather than retained, then detection latency for injected LLM/control-plane degradation will be higher.	L0/L1 IOL	HELIOS-noIOL + injection	Detection latency	BCa bootstrap CI	Exploratory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ3	**A-H4:** If MAHC uncertainty-aware consensus is replaced by vote-only aggregation (HELIOS-noConsensus) rather than retained, then HR@3 on the ambiguous-incident stratum will be lower.	L3 MAHC	HELIOS-noConsensus	HR@3 (Ambigu-ous)	Wilcoxon signed-rank	Ablation / Confirma-tory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ3	**A-H6:** If only the G-pipe is used (HELIOS-G) rather than HELIOS-Full, then HR@3 on narrative-heavy incidents will be lower.	L2b G-pipe	HELIOS-G	HR@3 (Narrative-heavy)	Wilcoxon signed-rank	Ablation / Confirma-tory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ3	**A-H7:** If the L-pipe is removed (HELIOS-noLLM) rather than retained, then factual and structural hallucination rates (H_fact, H_struct) will be at or near zero, versus greater than zero under HELIOS-Full.	L/G-pipes	HELIOS-noLLM (DV: hallucination)	H_fact, H_struct	Sign test	Ablation / Confirmatory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ3	**A-H8:** If UEG-C typed-edge composi-tion, subgraph-aware training, and Agent-Reasoning-Step nodes are disabled (HELIOS-noStructural) rather than retained, then HR@3 will be lower.	L1 UEG-C structural sub-mechanisms	HELIOS-noStructural	HR@3	Wilcoxon signed-rank	Ablation / Confirma-tory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ3	**E-H1:** If P4 cognitive nodes are removed (HELIOS-noCogni) rather than retained, then HR@3 on the Tier 4 agentic-AI subset will be lower.	L1 UEG-C cognitive nodes	HELIOS-noCogni	HR@3 (Tier 4 subset)	BCa bootstrap CI	Exploratory (reclassi-fied)

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ3	**E-H2:** If uniform routing is used (HELIOS-noRouter) rather than ORAR, then time-to-triage (TTT) will be higher.	L3 ORAR CBR	HELIOS-noRouter	TTT	BCa bootstrap CI	Exploratory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ3	**E-H3:** If the Action-Class Predictor is removed (HELIOS-noACP) rather than retained, then latency-constraint satisfaction will be lower.	L3 ACP	HELIOS-noACP	Latency-constraint satisfaction	BCa bootstrap CI	Exploratory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ3	**E-H4:** If FGSV is disabled (HELIOS-noFGSV) rather than retained, then the false-accept rate under synthetic fidelity-break perturba-tion will be higher.	L4 FGSV	HELIOS-noFGSV	False-accept rate	BCa bootstrap CI	Exploratory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ3	**E-H5:** If reward rec-onciliation is disabled (HELIOS-noReconcile) rather than retained, then Reward Reconcilia-tion Error on delayed-label incidents will be non-zero (and higher than under HELIOS-Full).	L4 Reconcile	HELIOS-noReconcile	RRE per incident	BCa bootstrap CI	Exploratory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ3	**E-H8:** If the Action-Class Predictor is removed (HELIOS-noACP) rather than retained, then latency-constraint satisfaction will be lower.	L3 ACP	HELIOS-noACP	Latency-constraint satisfaction	BCa bootstrap CI	Exploratory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ3	**E-H9:** If hardware isolation is reduced to HIS-3 (HELIOS-HIS3) rather than the design-committed isolation tier, then per-incident latency will be higher.	Engineering-validity	HELIOS-HIS3	Δ-latency per HIS tier	BCa bootstrap CI	Exploratory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ3	**E-H10:** If the C2 structural bundle is ablated (HELIOS-noStructural), then the ΔHR@3 will be attributable across typed-edge composi-tion, subgraph-aware training, and ARS-node sub-mechanisms.	L1 UEG-C structural sub-mechanisms	HELIOS-noStructural (descriptive sub-analysis)	ΔHR@3 attribution across typed-edge / training-discipline / ARS-node sub-mechanisms	Descriptive with BCa bootstrap CI	Exploratory

RQ	Hypothesis	Component	Variant	Primary Metric	Statistical Test	Family / Status
RQ4	**E-H7:** If Chain-of-Evidence narrative is shown rather than a raw ranked list (HELIOS-Full vs HELIOS-noExpl), then SRE correct-interpretation rate and trust will be higher and time-to-decision will be lower	L3 Narrative	HELIOS-noExpl vs. Full	Interp. rate; TTT; trust	Within-subjects study (paired Wilcoxon)	Exploratory

Note. McNemar's exact test is used for paired binary HR@3 outcomes in the baseline-comparison family (B-H1 through B-H5); Wilcoxon signed-rank is the binding test for paired continuous outcomes throughout (ablation family; B-H6, B-H7, B-H8). For B-H6, B-H7, and B-H8, paired t-tests on log-transformed outcomes are reported as sensitivity analysis only and are not the binding inference. Exploratory hypotheses receive bias-corrected and accelerated bootstrap confidence intervals but do not enter the Holm–Bonferroni family-wise correction. HR@3 in HELIOS-noConsensus is measured at the same Plane 4 architectural location as in HELIOS-Full, where the vote-only aggregator replaces the MAHC consensus rule but emits a structurally identical ranked-output tuple; the construct-validity asymmetry that this measurement-location identity does not dissolve is acknowledged in Section 3.9.2. C1 (runtime-enforced evaluation rigor) does not appear in this table because it is evaluated descriptively through verification-rate, completion-rate, and ledger-entry-count metrics (Section 3.6.8 and Table 1) rather than via a confirmatory or exploratory hypothesis.

For every contribution C1 through C6 with confirmatory hypothesis A-Hy or B-Hy, failure to reject the null hypothesis at the pre-registered Holm–Bonferroni-adjusted α implies lack of empirical support for the contribution under the conditions specified in this chapter; negative confirmatory outcomes do not invalidate the architectural specification of the contribution but bound the empirical evidence base, and would prompt a corresponding revision of the contribution's evidential status in the final dissertation. For exploratory hypotheses, no confirmatory inference is drawn; exploratory results inform but do not test the corresponding contribution claim and inform the design of future confirmatory pre-registrations. Specific to A-H4 (no-consensus on ambiguous incidents) and A-H8 (HELIOS-noStructural bundle), the per-hypothesis power audit in Section 3.8.3 indicates these tests are underpowered at the delivered corpus; the pre-registered inferential

rule for these two hypotheses is asymmetric — rejection at $\alpha = 0.00625$ supports the contribution; non-rejection is reported as "inconclusive at delivered power" rather than "no support." This asymmetric rule is locked at OSF freeze.

1.9.3 Statistical Methods

The statistical analysis will be structured around five methodological commitments, summarized here and detailed below: (i) within-subjects paired design with Wilcoxon signed-rank as primary test; (ii) family-wise error control through separate Holm–Bonferroni correction within each hypothesis family with pre-registered rank ordering; (iii) effect-size derivation grounded in minimum-detectable-effect framing rather than operational-significance midpoint; (iv) BCa bootstrap confidence intervals for every variant–benchmark–metric cell; (v) pre-registration discipline with deviation-log enforcement.

First, paired design. The experiment will be a within-subjects paired design: every fault instance will be processed by HELIOS and by each comparator variant or baseline, so each fault will act as its own control. The validity of the paired-comparison test relies on within-pair independence of treatment effects across incidents, which is operationally protected by the static-policy commitment documented in Section 3.3.3 and Section 3.6.5. The on-line LinUCB and ACP learning dynamics will be evaluated separately on the disjoint exploratory delayed-label subset and do not contribute to confirmatory inference. The primary test for paired non-parametric comparisons of HR@k, MTTR, and CpR across variant–benchmark cells will be the Wilcoxon signed-rank test, in one-sided form where a directional expectation is pre-registered and two-sided form otherwise. The rank-based mixed-effects analysis specified in Section 3.5 (with nparLD as concurrent sen-

sitivity check) is applied on Tier 1 only to verify pseudoreplication-controlled robustness of the Wilcoxon inference and does not supersede the primary confirmatory result.

Second, family-wise error control. Family-wise error rate will be controlled through Holm–Bonferroni correction applied separately within each hypothesis family with pre-registered rank ordering. The rank-1 adjusted α is 0.00625 for the eight-hypothesis baseline-comparison family and 0.00625 for the eight-hypothesis ablation family.

The Holm rank order is binding from OSF freeze and is established by a priori expected effect size, largest first, so that the most-likely-to-reject hypothesis is tested at the strictest α and the procedure terminates early if H_(1) fails to reject. The pre-registered rank order for the ablation family is: A-H3 (D-pipe statistical floor; expected ΔHR@3 ≥ 0.15) > A-H7 (hallucination zero without LLM; expected H_fact ≈ 0 vs > 0) > A-H1 (LLM necessity; expected ΔHR@3 ≥ 0.10) > A-H2 (LLM-only sufficiency; expected ΔHR@3 ≥ 0.10) > A-H6 (graph-only on narrative-heavy; expected ΔHR@3 ≥ 0.12) > A-H5 (uniform routing increases CpR; expected Δlog-CpR ≥ 0.60) > A-H4 (no-consensus on ambiguous; expected ΔHR@3 ≥ 0.05, underpowered) > A-H8 (HELIOS-noStructural bundle; expected ΔHR@3 ≥ 0.08, underpowered). The pre-registered rank order for the baseline-comparison family is: B-H1 through B-H5 (HELIOS exceeds five baselines on HR@3) ordered by expected discordance with HELIOS-favorable ratio, followed by B-H7 (cost reduction; expected δ = 0.80) > B-H6 (MTTR reduction; expected δ = 0.60) > B-H8 (Avg@5; expected ΔAvg@5 ≥ 0.10). Holm–Bonferroni is selected over plain Bonferroni because several hypotheses in each family are correlated. Family-wise error control under Holm–Bonferroni is governed by the number of hypotheses tested within a family, not by the sample size of the underlying paired observations; the symmetric rank-1 α across the two eight-hypothesis families is therefore the pro-

cedurally correct correction regardless of corpus-size asymmetry, with per-test power (rather than family-wise error) accounting for the corpus-size difference. Benjamini–Hochberg false discovery rate at q = 0.05 is reported as a sensitivity analysis; confirmatory inference is FWER-controlled.

Third, effect-size derivation under minimum-detectable-effect framing. The binding effect size is specified as the minimum effect detectable at the delivered corpus under the pre-registered statistical procedure, rather than as a researcher-chosen operational-significance threshold. At the delivered n = 174 paired observations, one-sided Holm–Bonferroni-adjusted α = 0.00625, paired correlation ρ = 0.5, and 80 percent power, the protocol detects a paired-difference effect of Cohen's h ≥ 0.276 (computed as h = 2·arcsin(√p□) − 2·arcsin(√p□) using IEEE-754 double precision, rounded to three decimal places). Against the AIOpsLab-equivalent CHASE baseline of HR@3 = 0.60 (derivation in Table 5 Panel A), this corresponds to a HELIOS target of HR@3 ≥ 0.73, or an absolute improvement of 13 percentage points.

The 13-percentage-point framing is presented as a consequence of the minimum-detectable-effect specification rather than as a researcher-chosen operational-significance midpoint. Industrial-deployment retrospectives (Shan et al., 2025; T. Wang & Qi, 2024) describe a qualitative noise-significance transition in production AIOps deployments where HR@3 improvements small enough to be obscured by operational noise produce no detectable downstream effect in MTTR or CpR telemetry, while improvements large enough to clear that noise floor produce persistent and measurable production effects; the protocol's minimum-detectable effect of 13 pp falls within the qualitative band described by these retrospectives, supporting the practical relevance of the detection threshold without grounding it numerically in the citations. Sensitivity is reported across h □ [0.213, 0.339] in Table 6 Panel B.

The CHASE-to-AIOpsLab adjustment factor that will translate CHASE's published HR@3 (0.72) into the AIOpsLab-equivalent baseline applies three adjustments: service-count dilution (−0.04 to −0.06), fault-taxonomy expansion (−0.02 to −0.04), and labeling-discipline strictness (−0.02 to −0.04), totaling −0.08 to −0.14 with midpoint −0.12 and yielding an AIOpsLab-equivalent CHASE baseline of HR@3 = 0.60. Each adjustment component, its citation source, and the sensitivity of the binding effect size to ±25 percent perturbation of each component will be reported in Table 5-8 and locked at OSF freeze.

Applying the minimum-detectable-effect framing to the adjusted baseline of HR@3 = 0.60 yields a HELIOS target of HR@3 = 0.73 and the corresponding Cohen's h = 0.276 (one-sided; arcsine-transformation paired difference). At one-sided $\alpha = 0.00625$, paired correlation $\rho = 0.5$, and 80 percent power, the required paired sample size will be $n = 73$ faults per pair; inflated 15 percent for attrition yields $n = 84$, and the delivered corpus of 174 will be 2.1× the statistical minimum. A sensitivity analysis is reported across paired correlations $\rho \in \{0.4, 0.5, 0.6\}$ and across $h \in [0.213, 0.339]$. Effect sizes for continuous outcomes are MTTR $\delta = 0.60$ (medium, on log-scale) and Cost $\delta = 0.80$ (large, consistent with FrugalGPT's reported 40–98 percent cost-reduction range; (L. Chen et al., 2023)). The corpus is robust to $h \geq 0.25$ but would be under-powered for $h < 0.20$; this bound is reported explicitly as a scope condition (Table 4).

A per-hypothesis power audit across the eight ablation hypotheses at $\alpha = 0.00625$, $\rho = 0.5$, and $n = 174$ will be reported in Table 7 Panel C. A-H1, A-H2, A-H3, A-H5, A-H6, and A-H7 all meet or exceed 80 percent power at the delivered corpus. A-H4 (ambiguous-incident consensus, $\Delta HR@3 \geq 0.05$, $h \approx 0.10$ against an adjusted baseline of 0.55) is under-powered at approximately 65 percent. A-H8 (structural-and-integration bundle, $\Delta HR@3 \geq 0.08$, $h \approx 0.18$ against the adjusted AIOpsLab baseline of 0.60) is under-powered at approximately 62 percent. Both A-H4 and A-H8

are retained in the confirmatory family with the pre-registered asymmetric inferential rule specified in Table 3 footnote: rejection at α = 0.00625 supports the contribution; non-rejection is reported as "inconclusive at delivered power" rather than "no support."

Panel D reports the per-hypothesis power audit for the baseline-comparison family under McNemar's exact test (B-H1 through B-H5) and Wilcoxon signed-rank (B-H6 through B-H8). McNemar's power depends on the discordant-pair proportion p_disc = p□□ + p□□ and the discordance ratio r = p□□ / (p□□ + p□□), not on Cohen's h directly; the audit is therefore presented across (p_disc, r) cells. At one-sided α = 0.00625 and 80 percent power, the delivered n = 174 supports confirmatory inference for B-H1–B-H5 at p_disc ≥ 0.30 with r ≥ 0.70 (HELIOS-favorable). Below this threshold the family is under-powered; the asymmetric inferential rule applies to any B-Hx whose realized discordance falls below the powered range.

Table 7
Sample-Size Sensitivity for Cohen's h under One-Sided α = 0.00625, ρ = 0.5, 80% Power

Cohen's h	Required n per pair (no attrition)	Required n per pair (15% attrition)	Adequacy of delivered n = 174
0.15	≈ 245	≈ 282	Under-powered
0.20	≈ 138	≈ 159	Marginal
0.276 (binding)	**73**	**84**	**2.1× minimum**
0.35	≈ 47	≈ 54	**3.2× minimum**

Note. Sample sizes are computed under the paired-difference Cohen's *h* approximation for arcsine-transformed proportion differences.

Table 8
Panel A — CHASE-to-AIOpsLab Adjustment Components

Adjustment Component	Lower Bound	Midpoint (Binding)	Upper Bound	Source
Service-count dilution (CHASE → AIOpsLab)	−0.04	−0.05	−0.06	(L. Wu et al., 2020; Pham et al., 2024a; X. Zhang et al., 2025)
Fault-taxonomy expansion (4 → 6 categories)	−0.02	−0.03	−0.04	AIOpsLab specification (Y. Chen et al., 2025)
Labeling-discipline strictness (multi-cause → single-cause)	−0.02	−0.03	−0.04	AIOpsLab specification (Y. Chen et al., 2025); CHASE specification (Zhao et al., 2025)

Adjustment Component	Lower Bound	Midpoint (Binding)	Upper Bound	Source
Total adjustment	**−0.08**	**−0.12**	**−0.14**	—
Adjusted CHASE-equivalent baseline on AIOpsLab	**HR@3 = 0.64**	**HR@3 = 0.60**	**HR@3 = 0.58**	Derived

Table 9
Panel B — Operational-Significance Threshold Sensitivity

Threshold Position	Threshold Value	HELIOS Target HR@3	Cohen's h	Required n per Pair (15% Attrition)	Adequacy at Delivered n = 174
Lower bound (−25%)	10 pp	0.70	0.213	145	Marginal (1.2× minimum)
Midpoint (binding)	**13 pp**	**0.73**	**0.276**	**84**	**Adequate (2.1× minimum)**

Threshold Position	Threshold Value	HELIOS Target HR@3	Cohen's h	Required n per Pair (15% Attrition)	Adequacy at Delivered n = 174
Upper bound (+25%)	16 pp	0.76	0.339	56	Robust (3.1× minimum)

Table 10
Panel C — Per-Hypothesis Power Audit (Ablation Family, α = 0.00625, ρ = 0.5, n = 174)

Hypothesis	Effect-Size Target	Effect Size (h or δ)	Adjusted Baseline	Power at n = 174	Status
A-H1 (LLM necessity)	ΔHR@3 ≥ 0.10	$h \approx 0.22$	0.60	~85%	Adequate
A-H2 (LLM-only suffi-ciency)	ΔHR@3 ≥ 0.10	$h \approx 0.22$	0.60	~85%	Adequate

Hypothesis	Effect-Size Target	Effect Size (h or δ)	Adjusted Baseline	Power at n = 174	Status
A-H3 (D-pipe statistical floor)	ΔHR@3 ≥ 0.15	$h \approx 0.32$	0.60	~96%	Robust
A-H4 (ambiguous-incident consensus)	ΔHR@3 ≥ 0.05	$h \approx 0.10$	0.55	~65%	**Under-powered (disclosed)**
A-H5 (uniform routing CpR)	Δlog-CpR ≥ 0.60	δ = 0.60	—	~90%	Adequate
A-H6 (graph-only on narrative-heavy)	ΔHR@3 ≥ 0.12	$h \approx 0.27$	0.60	~91%	Adequate

Hypothesis	Effect-Size Target	Effect Size (h or δ)	Adjusted Baseline	Power at n = 174	Status
A-H7 (hallucination zero w/o LLM)	H_fact ≈ 0 vs. > 0	Sign test on near-zero rate	—	~93% (sign test, p□ = 0.05, p□ = 0.30, n = 174)	Adequate
A-H8 (structural-and-integration bundle)	ΔHR@3 ≥ 0.08	*h* ≈ 0.18	0.60	~62%	Under-powered (disclosed)

Note: A-H7 power is computed under the exact sign test for a one-sided difference in proportions (H_fact rate) at α = 0.00625, n = 174; the h-based formulation is not applicable because the null is a near-zero rate.

Table 11
Panel D — Per-Hypothesis Power Audit, B-Family under McNemar's Exact (α = 0.00625, 80% power)

(p_disc, r)	n Required	Adequacy at Delivered n = 174
(0.20, 0.60)	~530	Severely under-powered
(0.20, 0.70)	~325	Under-powered
(0.20, 0.80)	~200	Marginal
(0.30, 0.60)	~360	Under-powered

(p_disc, r)	n Required	Adequacy at Delivered n = 174
(0.30, 0.70)	~218	Marginal
(0.30, 0.80)	~140	Adequate (1.2× minimum)
(0.40, 0.60)	~270	Under-powered
(0.40, 0.70)	~165	Adequate (1.05× minimum)
(0.40, 0.80)	~108	Robust (1.6× minimum)

Note. Computed via mid-p McNemar's exact under one-sided alternative. The realized (p_disc, r) for B-H1 through B-H5 will be reported in Chapter Four; if any falls below the powered range, the asymmetric inferential rule applies.

The relationship between the Wilcoxon signed-rank test and the rank-based mixed-effects analysis specified in Section 3.5 requires explicit binding to eliminate researcher-degrees-of-freedom risk. The Wilcoxon signed-rank test on paired (variant, HELIOS-Full) differences within each (benchmark, metric) cell will be the binding confirmatory inference for the ablation family; the rank-based mixed-effects analysis (with problem_id as a random intercept and benchmark_source as a fixed effect, supplemented by nparLD) will be reported only on Tier 1 as a pseudoreplication-controlled sensitivity analysis whose role is to verify that the Wilcoxon inference does not shift materially when the within-Tier-1 dependence structure is modeled explicitly. For the baseline-comparison family, McNemar's exact test will be the binding confirmatory inference for paired binary HR@3 outcomes (B-H1 through B-H5); Wilcoxon signed-rank will be the binding confirmatory inference for paired continuous outcomes (B-H6, B-H7, B-H8). Paired one-sided t-tests on log-transformed continuous outcomes will be reported as sensitivity analyses

for B-H6, B-H7, and B-H8 but will not constitute the binding inference for any hypothesis, removing the conditional-test design that would inflate Type I error under Shapiro–Wilk gating. The rank-based mixed-effects analysis is not used as the confirmatory test for any baseline-comparison hypothesis. The binding is locked at OSF freeze; any post-hoc substitution would be recorded as a deviation-log entry with its analytic consequence — typically the affected hypothesis moving from confirmatory to exploratory status.

Fourth, BCa bootstrap confidence intervals. Confidence intervals will be reported for every variant–benchmark–metric cell using the bias-corrected and accelerated (BCa) bootstrap with 10,000 resamples; resampling is at the paired-incident level, the BCa seed sequence is pre-registered, and the acceleration estimate uses jackknife acceleration. Cliff's δ accompanies every Wilcoxon result; Cohen's d accompanies every paired-t result; Hodges–Lehmann estimates of location shift accompany every paired-difference reporting.

Fifth, pre-registration discipline. All confirmatory tests will operate under the pre-registration discipline specified in Section 3.5 and Section 3.3.3: pre-registered seed register, pre-registered SHA-256-hashed fault corpus, pre-registered variant feature-flag matrix bound to the binary commit, and pre-registered price book. Deviations will be recorded in the deviation log with timestamp, the clause deviated from, the concrete change, the reason, and the analytic consequence (typically: the affected test moves from confirmatory to exploratory). The deviation log is an append-only, cryptographically-signed file deposited at OSF at every commit; write access is restricted to the candidate, read access is public via OSF, and entries are written before any analysis benefiting from the deviation is conducted.

1.9.4 Evaluation Metrics

The confirmatory evaluation employs five primary metrics. Hit Rate at k (HR@k) for k □ {1, 3} is the fraction of incidents for which the ground-truth root cause appears in the top-k of the ranked root-cause list published by the MAHC consensus; HR@3 is the primary outcome for the ablation family and the baseline-comparison family. Mean Time to Resolution (MTTR), for incidents that receive autonomous remediation, is the elapsed time from anomaly detection to post-remediation telemetry restoration within the incident window [t_injection, t_injection + 1800 s]; it will be log-transformed prior to parametric analysis to correct for right-skew. Cost per Resolution (CpR) is the five-component sum of token cost, infrastructure cost, storage cost, human-minute cost, and opportunity cost, computed per incident under the pre-registered price book. Total CpR is the binding metric for B-H7; per-component decomposition will be reported descriptively (without confirmatory inference on individual components) to disambiguate algorithmic cost contributions from downstream-MTTR cost contributions, since opportunity cost (revenue forgone during MTTR) typically dominates the dollar-magnitude scale and could otherwise obscure token- and infrastructure-cost effects from algorithmic mechanisms. Time to Triage (TTT) is the elapsed time from anomaly detection to publication of the MAHC PROVISIONAL verdict. Hallucination rate is decomposed into structural (H_struct) and factual (H_fact) components, with independent-rater labeling against pre-specified rubrics and Cohen's $\kappa \geq 0.7$ required for inter-rater agreement.

Additional descriptive metrics — HR@10, mean reciprocal rank (MRR), mean first rank (MFR), precision, recall, macro F1, HITL escalation rate, sampling-regime stratification, hardware-isolation-tier stratification, retraining overhead in CPU-hours and GPU-hours, ethics-compliance rate, and the Reward Reconciliation Error on the pre-registered 25 percent delayed-label subset —

will be reported alongside the confirmatory set. The FAE-Score for narrative explanation quality, computed as the arithmetic mean across three judge models (GPT-4o, Claude Sonnet, and Gemini Pro) with 10 percent of judgments human-anchored for calibration, will accompany the Research Question 4 user study. These descriptive metrics support interpretability and secondary exploratory analysis; they are not subject to pre-registered hypothesis tests and do not enter the Holm–Bonferroni correction family.

1.10 Validity and Reliability

1.10.1 Internal Validity

Eight internal-validity threats are identified, each addressed by an explicit architectural or procedural mitigation that substantially strengthens — but does not eliminate — the corresponding internal-validity exposure.

Threat 1 — Input drift between variants. Because each variant will be produced from the same HELIOS binary by feature-flag toggling, and the snapshot hash registry will enforce that every active pipeline consumes the same content-hashed UEG-C snapshot for the same (fault, seed) pair, inter-variant variance arises only from the pipelines themselves. The paired-observation design will satisfy its structural precondition under this architectural enforcement.

Threat 2 — Hidden coupling between feature flags. The original `p4_cognitive` flag silently removed cognitive nodes from G-pipe's input; the resolution scopes the flag to L-pipe visibility through Persona-Profile tool-permission restriction. A hidden-coupling audit conducted at design freeze confirms that the remaining eleven flags toggle disjoint code paths. The audit certifies

disjoint code paths under single-flag toggling; it does not certify non-interacting semantic effects under arbitrary flag combinations. The chapter's confirmatory protocol ablates each contribution against HELIOS-Full singly, not in combination, so the joint-inactivation interaction surface is not exercised in the confirmatory family. Interaction-effect probes across selected flag combinations are identified as a direction for follow-on exploratory analysis.

Threat 3 — Cross-incident learning dependency in ORAR. This would violate the within-pair independence assumption if LinUCB parameters were updated on-line during the confirmatory protocol. The mitigation is the static-policy commitment (Section 3.3.3 and Section 3.6.5): LinUCB and ACP parameters will be frozen at the pre-warmed state for the entire confirmatory protocol, on-line updates restricted to the disjoint exploratory delayed-label subset, and freezing enforced at the binary level by the Variant Control Layer. The static-policy commitment narrows confirmatory inference for C6 to the cost-aware static-routing mechanism; on-line learning dynamics are evaluated only as exploratory evidence.

Threat 4 — LLM hallucination biasing the L-pipe verdicts. Hallucination is addressed at three architectural layers: at the telemetry layer, the P4 cognitive pillar captures hallucination candidates through tool-call schema-validation failures and through semantic drift between chain-of-thought and structured output; at the reasoning layer, the Cognitive Sentinel fires on detected pathological patterns; and at the consensus layer, the MAHC entropy regulariser tests for overconfident unanimous agreement and escalates to HITL rather than auto-executing.

Threat 5 — Reward hacking in the ORAR learning loop. Relevant only to the exploratory subset where on-line learning is enabled. Three anti-hacking regularisers will be applied during exploratory analysis: a variance penalty across similar incidents, an action-diversity bonus, and a held-out audit in which 5 percent of incidents are double-diagnosed by an independent agent for reward validation. Reward hacking is not a confirmatory threat because the confirmatory protocol freezes the learning loop.

Threat 6 — Control-plane failure during evaluation. When the Internal Cognitive Sentinel detects degradation in HELIOS's own LLM agents, the Deterministic Fallback Agent will activate using hand-coded SOPs for approximately 30 percent of common fault classes; remaining fault classes will degrade to alerting-only mode pending HITL intervention. The system cannot silently fail during evaluation, and control-plane degradation events are themselves logged as first-class covariates through the IOL mechanism.

Threat 7 — Adversarial input during telemetry ingestion. Adversarially-crafted input is contained by the Privacy Gateway and the Adversarial Robustness Monitor. Adversarial events are recorded as first-class CoE entries; incidents involving adversarial events will be stratified in analysis and will not contribute to confirmatory inference on the unperturbed cells.

Threat 8 — Opacity of the pre-warmed state. Discussed extensively in Section 3.3.3, where the disclosure regime — pre-registration of construction parameters, hash of the construction script, hash-verified disjointness at the (fault-type, service-family, injection-parameter) granularity, content-addressable parameter artifacts, runtime hash verification by the Variant Control

Layer — is specified. The residual exposure has two named components: structural similarity at coarser conceptual levels (shared HTTP semantics, anomaly-class taxonomy, OTEL conventions) that hash-disjointness at the instance level cannot exclude; and the construction-validity threat that the synthetic testbed and construction script were designed by the same researcher who designed HELIOS. The residual structural-similarity exposure is bounded by the small dimensionality of the LinUCB feature space (approximately 12 to 18 features, comprising incident features and the ACP stateful-action feature block) relative to the diversity of the confirmatory corpus, but neither this bound nor the disclosure regime fully dissolves the threat. External replication on a structurally distinct microservice paradigm — for example, an event-driven serverless mesh in place of a request-response Kubernetes deployment — would substantially strengthen the warrant beyond what the present disclosure regime achieves, and is identified in Section 3.11.5 as a priority direction for follow-on research.

1.10.2 Construct Validity

The primary construct-validity reliability assurance mechanism is architectural traceability: every construct C1 through C6 corresponds to a named architectural component, every independent variable is instantiated through a feature flag at the Variant Control Layer (or, for C1, through architectural commitment to evaluation-rigor infrastructure), and every dependent variable is measured at a specific architectural location with specified instrumentation (Section 3.3.6). This traceability substantially strengthens the reliability of construct measurement relative to comparator AIOps evaluations, without claiming to strengthen the validity of the operationalization in the stronger theoretical sense; the choice of architectural binding is itself a theoretical choice that determines which construct the measurement reflects.

Three secondary construct-validity threats remain. First, FAE-Score judge-model instability is mitigated by the three-judge ensemble plus periodic human-anchored calibration at 10 percent sample. Second, the SRE Likert trust scale in the RQ4 user study is a self-reported measure potentially subject to social-desirability bias; it will be triangulated with the objective HITL rejection counter, which is architecturally logged and cannot be modified by participants. Third, the Tier 4 cognitive-agentic faults used in the C2 exploratory empirical validation are constructed by the researcher specifically through NIKA-style instrumentation to exercise cognitive-node reasoning; they may not reflect the population of cognitive-agentic faults that would arise independently in deployed agentic systems. This threat motivates the reclassification of the C2 cognitive-agentic test as exploratory under E-H1 (Section 3.6.8 and Section 3.8.2).

1.10.3 External Validity

External validity is addressed at three levels. First, the evaluation will span five public benchmarks with heterogeneous microservice architectures and fault classes (AIOpsLab, DeathStarBench, Train-Ticket, PetShop, OpsEval), providing breadth of coverage to the extent that the corpus distribution permits. The 174-fault corpus is concentrated in the AIOpsLab benchmark family (approximately 76 percent from Tier 1 and Tier 2; Section 3.5). Confirmatory external-validity claims are therefore bounded to AIOpsLab and its parametric extension. Cross-benchmark results from Tier 3 (Train-Ticket and DeathStarBench under ChaosBlade injection) and Tier 4 (synthetic cognitive-agentic faults layered on AIOpsLab) are positioned as external-validity probes rather than as confirmatory cross-benchmark generalization: they will suggest that observed effects extend beyond the AIOpsLab family, but the per-benchmark sample sizes are insufficient for adequately powered between-benchmark inference. Generalization to environments outside AIOpsLab and its

parametric extension — for example, environments instrumented with non-OTEL telemetry, with substantially larger service counts, with proprietary observability stacks, or operating under fault regimes absent from the AIOpsLab fault taxonomy — would require an industrial replication study beyond the scope of this dissertation.

Second, the four-tier corpus augmentation provides triangulation across injection tools (Chaos Mesh, ChaosBlade, NIKA-style instrumentation) and fault taxonomies, supporting the secondary claim that observed effects will not be artifacts of a single fault-injection tool's failure mode; it does not support a claim of population-level generalization. Third, HELIOS's five core assumptions A1 through A5 (Section 3.11.1) bound the population to which findings generalize; assumption violations are disclosed rather than hidden. The external-validity claim of the study is therefore that confirmatory external validity holds for AIOpsLab and its parametric extension under the five core assumptions, with cross-benchmark results in Tiers 3 and 4 reported as external-validity probes that motivate, but do not constitute, broader generalization.

1.10.4 Reliability

Reliability will be assured through four commitments. First, all primary hypotheses will be pre-registered at OSF with a time-stamped freeze that binds the analysis plan; the freeze date and OSF deposit DOI will be recorded in the deviation log alongside this chapter. Second, all benchmark datasets, baseline-system configurations, and fault-instance hashes will be published at pre-registration, supporting independent replication. Third, the artifact specification in this chapter is at a level of detail sufficient for independent replication, and the feature-flag matrix, the algorithm specifications, and the metric-integrity-gate specification are binding contracts. Fourth, the CoE ledger's append-only, cryptographically-signed, Conflict-free Replicated Data Type (CRDT)-

replicated design will ensure that decision records produced during evaluation are auditable and tamper-evident.

Reproducibility will be pipeline-scoped, protocol-scoped, and tiered, in accordance with the bounded-reproducibility principle specified in Section 3.6.2. D-pipe and G-pipe will be reproducible to the level of bitwise output equality under the binding hardware and software configuration documented in Section 3.6.2. L-pipe will be reproducible at the level of greedy-decoded token-sequence determinism under Protocol A (Llama-3.1-70B-Instruct served via vLLM 0.6.x with `enforce_eager=True`, `disable_custom_all_reduce=True`, single-replica serving, temperature = 0, top-p = 1, top-k = 1); container digest, weights SHA-256, and tensor-parallel-degree configuration are deposited at OSF. Logit-level variance ε under Protocol A will be empirically measured on a 200-prompt probe set during the Chapter Four pilot pass and reported in Chapter Four with its standard error and measurement date; the measurement is conducted prior to chapter submission so that ε is a delivered empirical result at viva. Protocol B (commercial API access) will be reported as a sensitivity analysis. Any variant replicate that fails token-sequence equality under Protocol A will be logged as a deviation-log event and will not contribute to confirmatory inference.

A model-selection sensitivity analysis will defend against the threat that Protocol A results would not replicate under an alternative open-weight model family. A subset of the confirmatory ablation protocol will be repeated on Llama-3.1-70B-Instruct (binding) and Qwen-2.5-72B-Instruct (binding contrast family). Qwen-2.5-72B-Instruct is selected as the contrast family rather than Mixtral-8x22B-Instruct because its dense decoder-only architecture and 72-billion parameter count are closer to Llama-3.1-70B-Instruct's architecture and parameter count than Mixtral-8x22B's mixture-of-experts design, providing a more comparable contrast for the pattern-of-effects

invariance question. Both model identifiers will be content-hashed and deposited at OSF. Absolute scores are not expected to be invariant across families; the sensitivity analysis is designed to demonstrate that the ablation conclusions will not be artifacts of a single model choice.

Every artifact-identifier reference in this chapter — the binding hardware configuration, the binding model identifiers, the container digests, the frozen LinUCB and ACP parameter artifacts, the synthetic pre-training corpus and its construction script, the deviation-log file, and the GLMM specification — will be deposited at OSF as a content-addressable artifact at chapter submission. The Variant Control Layer will verify the hash of each loaded artifact against the hash recorded at OSF freeze at every confirmatory run start; hash divergence will block execution.

1.11 Ethical Considerations

The research will be conducted under the ethical standards of the researcher's institutional review board, applicable data-protection regulations, and the responsible-AI-research practices appropriate to a study whose artifact has potential autonomous-action capability.

Telemetry data handling is governed by the Privacy Gateway at the Data-Plane-to-Control-Plane boundary. Pattern-based PII detection and named-entity recognition for free-text log content will redact sensitive tokens before any signal reaches the Control Plane; redacted tokens will be replaced with deterministic placeholders that preserve grammatical structure for downstream LLM parsing. Per-tenant differential-privacy budgets will cap aggregated-query information leakage, and multi-tenant isolation within the UEG-C is enforced at the graph-partition level. Raw telemetry collected in any industrial-replication extension will remain within the industrial partner's infrastructure; only anonymized aggregate metrics will cross institutional boundaries.

The ethics of autonomous AI decision-making are addressed architecturally. The Monitor Agent will enforce ethics constraints across every automated action and record ethics-audit results as first-class CoE entries. The dynamic autonomic threshold α classifies all actions into risk tiers; high-risk actions (database modifications, network-policy changes, rollbacks) will unconditionally require human approval via the HITL SRE Gate. No autonomous remediation will be executed against production telemetry without prior validation in the FGSV shadow twin. These mechanisms are architectural first-class components, not post-hoc compliance additions.

Human participants in the RQ4 user study and any industrial-replication case study will participate voluntarily, with full disclosure of the study's purpose, the nature of the AI system under evaluation, and the data collected. Informed consent will be obtained in writing prior to any data collection, and participants may withdraw at any time without penalty. Participant data (Likert responses, override counts, time-to-decision measurements) will be anonymized at collection; only aggregate statistics will enter the dissertation. The user-study protocol, recruitment materials, consent forms, and debriefing script (Section 3.7) will be submitted to the researcher's institutional ethics review board for approval prior to data collection; IRB approval status will be recorded in the deviation log at OSF freeze. Participant data will be stored in accordance with applicable regulations including GDPR where relevant.

Responsible AI research practice is operationalized at the level of the artifact design itself: the Monitor Agent, HITL gate, Adversarial Robustness Monitor, Privacy Gateway, and FGSV shadow validator are architectural first-class components rather than review-stage compliance additions. Who sets the risk threshold, who may approve a high-risk change, and who reviews overrides and spend is specified in the business evaluation layer. Legal liability for a bad automatic

change remains outside the study. Power asymmetries that would follow from running the artefact at scale, beyond those decision rights, remain a direction for follow-on work.

1.12 Limitations of the Methodology

1.12.1 Core Assumptions and Their Scope Conditions

Five core assumptions bound the scope of findings; where they are violated, conclusions should be applied with caution.

A1 — OTEL instrumentation. Target microservices and agent platforms are instrumented (or instrumentable via sidecars) to emit OTEL-compliant signals. Performance in partially-instrumented environments is not evaluated.

A2 — Dedicated control-plane compute. An enterprise cluster overhead budget of 5 to 10 percent is available for the HELIOS Control Plane. HELIOS can operate in reduced-capability mode for resource-constrained environments, but the full multi-pipeline architecture requires enterprise-scale resources and reduced-capability mode is not the primary subject of evaluation.

A3 — Initial accuracy-calibration data. Bootstrap accuracy priors for the MAHC hierarchical-prior mechanism and the ORAR LinUCB arm parameters are seeded from public benchmark performance and pre-production fault-injection campaigns. Findings should not be extrapolated to organizations without a 30-day HITL calibration phase.

A4 — LLM access tier. Full P4 cognitive telemetry — a primary component of Contribution C2 — requires at minimum Tier 2 (logprob-API) LLM access. Tier 3 deployments receive reduced cognitive signal quality and P4-dependent findings may not replicate under Tier 3 conditions.

A5 — Historical incident data. The Knowledge Cortex warm tier requires seed incident records for retrieval-augmented-generation-based SOP matching. Cold-start organizations face reduced SOP-matching accuracy for approximately 90 days post-deployment.

1.12.2 Corpus and Evidential-Status Limitations

L1 — Benchmark concentration. Approximately 76 percent of the 174-fault corpus is drawn from the AIOpsLab benchmark family (Section 3.5). Confirmatory inferences are bounded to AIOpsLab and its parametric extension (Section 3.9.3); cross-benchmark results in Tiers 3 and 4 will be reported as external-validity probes rather than as confirmatory cross-benchmark generalizations. Generalization to non-AIOpsLab environments requires an industrial replication study identified as a priority direction for follow-on research.

L2 — Cognitive-agentic fault evaluation. The empirical evaluation of Contribution C2 on agentic-AI fault classes is constrained by the synthetic construction of the Tier 4 corpus and the limited sample size (n = 12), which is below the 73-fault statistical minimum required for adequately powered confirmatory inference. Tier 4 instances are constructed by the researcher through NIKA-style instrumentation specifically to exercise cognitive-node reasoning, raising a construct-validity concern that Tier 4 may not sample a population of cognitive-agentic faults arising independently

outside the construction protocol. The Tier 4 substrate is layered onto AIOpsLab rather than constructed independently, so the E-H1 evidence base is bounded to "AIOpsLab plus cognitive-agentic instrumentation"; an externally constructed cognitive-agentic fault corpus on a non-AIOpsLab substrate is the priority direction for upgrading E-H1 to confirmatory status.

L3 — Architectural contributions with exploratory empirical validation. Contributions C3′ (FGSV) and C5 (IOL) are positioned as architectural contributions with exploratory empirical validation rather than as confirmatorily evidenced contributions on the present corpus. The empirical validation of FGSV requires synthetic fidelity-break perturbations and the empirical validation of IOL requires injected control-plane degradation, neither of which arises spontaneously from the public benchmarks. Future confirmatory pre-registrations with pre-specified effect sizes for these synthetic perturbation protocols are identified as the appropriate path to upgrading C3′ and C5 from exploratory to confirmatory empirical evidence.

1.12.3 Technical Limitations

Five technical limitations further constrain the scope of claims. HELIOS will treat each incident as independent; cascade failures manifesting as multiple simultaneous alerts will spawn parallel diagnostic episodes rather than a single correlated investigation, with multi-incident correlation scoped as future work. The stratified fault-injection curriculum covers known fault classes but does not provide a formal coverage measure over the unknown fault surface; few-shot adaptation claims are conditional on curriculum coverage of the target fault class. The FGSV ASRE WAL-replay mode introduces 5 to 30 seconds of additional latency for stateful remediations depending on WAL size; reported MTTR for stateful fault classes will reflect this overhead. Hot-reloadable

Persona Profiles create a governance race condition during active incident episodes; profile refresh is restricted to episode boundaries. The FAE-Score multi-judge ensemble mitigates single-judge bias but remains subject to judge-model drift; periodic human-anchored calibration at 10 percent sample detects drift but cannot prevent it.

1.12.4 Methodological Trade-offs

The research design embeds seven principal methodological trade-offs, each with a documented implication for the scope of findings. Latency versus accuracy is resolved by Fast-Path PROVISIONAL at 30 seconds and CONFIRMED on full quorum, with MTTR including PROVISIONAL time. Cost versus quality is resolved by ORAR cascade plus ACP plus Non-LLM-First Gate; CpR results are configuration-dependent. Twin fidelity versus speed is resolved by the three-mode ASRE; MTTR for stateful incidents exceeds MTTR for routing incidents. Coverage versus overhead is resolved by Tiered Adaptive Sampling with backpressure-aware downgrade; under extreme resource pressure rare brief faults may be missed. Autonomy versus safety is resolved by the dynamic α guardrail and mandatory HITL for high-risk actions, with HITL latency not experimentally controlled. Generalization versus depth is resolved by five benchmarks with eight baselines and a priori power analysis; class-specific performance variance may exceed aggregate effect-size estimates. *Confirmatory rigor versus on-line learning* is resolved by the static-policy commitment for ORAR (Section 3.3.3, Section 3.6.5): the confirmatory protocol freezes LinUCB and ACP parameters, narrowing confirmatory inference for C6 to the cost-aware static-routing mechanism, while on-line learning dynamics are reserved for exploratory analysis.

1.12.5 Scalability and External-Replication Boundaries

The 150-millisecond Plane 2 latency SLO and the 10,000-edge per-incident subgraph bound are design commitments validated on microservice estates of at most approximately 100 services within the benchmark regime. Scaling to 500, 1,000, 2,500, and 5,000 synthetic services is a pre-registered separate experiment; the scalability claim is to be earned on measured results, not asserted as an architectural property. Multi-region deployments require CRDT CoE replication and may require hierarchical Knowledge Cortex caching to maintain query latency.

The external validity of qualitative and longitudinal findings would be constrained by the use of a single industrial replication site, if such a site is engaged. Findings specific to organizational dynamics of adoption (SRE trust trajectories, override-rate decline patterns, integration effort) may not generalize to organizations with substantially different operational cultures, service portfolios, or AI maturity levels. Replication studies across multiple industrial sites — and external replication of the static-policy pre-warming pass on a structurally distinct microservice paradigm (for example, an event-driven serverless mesh in place of a request-response Kubernetes deployment) — are identified as priority directions for follow-on research. These are the principal interventions that would substantially strengthen the warrant for the construction-validity threat (Section 3.3.3, Threat 8 of Section 3.9.1) beyond what the present disclosure regime achieves.

1.13 Summary

This chapter has specified the methodology of a Design Science Research study whose central methodological contribution is the runtime enforcement of pre-registration discipline through architectural mechanisms. The Variant Control Layer, snapshot hash registry, metric integrity gate,

exclusion ledger, deviation log, and reconciliation ledger collectively constitute Contribution C1 — a research-method-as-system-architecture contribution that transforms DSR evaluation rigor from a reporting convention into a runtime invariant of the artifact, and that is generalizable beyond HELIOS to AIOps research-method practice and arguably to DSR research-method practice more broadly.

Within this methodological frame, six named contributions are advanced: C1 (runtime-enforced evaluation rigor), C2 (UEG-C with cognitive nodes), C3′ (FGSV evaluation-infrastructure discipline), C4 (MAHC uncertainty-aware consensus), C5 (IOL meta-observability), and C6 (ORAR cost-aware adaptive routing with cross-field-transfer profile from delayed-feedback bandits). The five design principles — pipeline independence, ablation-first design, bounded reproducibility, separation of diagnosis from action, and failure-as-covariate discipline — are intended as transferable design knowledge contributions in the (Gregor & Hevner, 2013) sense.

The research philosophy grounds the work in critical realism with a pragmatist orientation toward artifact evaluation. HELIOS is specified as an experimental artifact at a level of detail sufficient to justify the construct-to-variable mapping, the data-collection instruments, and the planned hypothesis tests. The data-analysis plan pre-registers two confirmatory hypothesis families corrected separately under Holm–Bonferroni with pre-registered rank ordering — an eight-hypothesis ablation family testing the internal contribution of each architectural mechanism, and an eight-hypothesis baseline-comparison family testing external comparative advantage — together with a ten-hypothesis exploratory family. Effect sizes are framed under minimum-detectable-effect logic at the delivered corpus: the protocol detects Cohen's $h \geq 0.276$ at $n = 174$, one-sided $\alpha = 0.00625$, $\rho = 0.5$, 80 percent power, corresponding to a 13 percentage-point HR@3 improvement against the AIOpsLab-equivalent CHASE baseline of 0.60.

The central methodological commitment is that the architecture and the evaluation will be coupled by design under documented constraints rather than by-construction guarantees. The Class 1 evaluation-rigor apparatus — the components that constitute Contribution C1 — substantially strengthens the warrant for confirmatory inference under the bounded scope conditions of the study without claiming to dissolve every threat to validity. The chapter is honest about residual exposures: the construction-validity threat from researcher-designed pre-warming corpus is bounded but not dissolved; the AIOpsLab corpus concentration bounds external validity to AIOpsLab and its parametric extension; the n = 12 Tier 4 cognitive-agentic corpus supports an existence-proof for C2 cognitive-agentic claims rather than a population-efficacy claim; and A-H4 and A-H8 are confirmatorily retained under an asymmetric inferential rule that treats non-rejection as 'inconclusive at delivered power' rather than 'no support.' Each residual exposure is mapped to a priority follow-on study identified in Section 3.11. Chapter Four will present the evaluation results obtained under this methodology.

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